



香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

ELEC 2100 Signals and Systems

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Outline Chapter 2 Time domain analysis of LTI systems



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1 Overview

2 System Differential Equations (for reference)

3 Continuous-Time System Response

4 Continuous-Time Convolution

5 Difference Equations (for reference)

6 Unit Impulse Response

7 Discrete Convolution

1. Overview of Time Domain Analysis Methods

- **Input/Output Description: Single-variable n-th-order differential (difference) equation**
- **State Variable Description: n-variable first-order differential (difference) equations**

Directly solving system differential/difference equations without transformations offers:

- Intuitive understanding
- Clear physical interpretation
- Foundational basis for learning transform-domain methods

Chapter 2 first introduces the Input/Output description method

1. Overview of Time Domain Analysis Methods

(1) What should we do if the system becomes more complex?

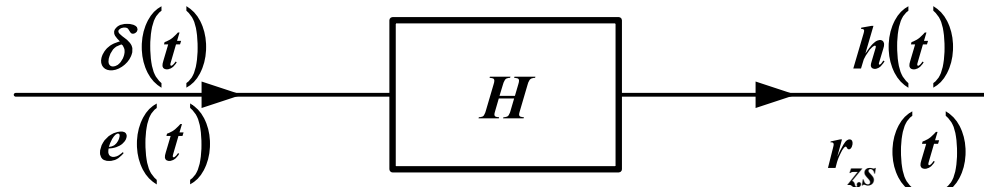
$$\begin{aligned} & a_n \frac{d^n r(t)}{dt^n} + a_{n-1} \frac{d^{n-1} r(t)}{dt^{n-1}} + \dots + a_1 \frac{dr(t)}{dt} + a_0 r(t) \\ &= b_m \frac{d^m e(t)}{dt^m} + b_{m-1} \frac{d^{m-1} e(t)}{dt^{m-1}} + \dots + b_1 \frac{de(t)}{dt} + b_0 e(t) \end{aligned}$$

(2) What should we do if the signal becomes more complex?

- DC, Sine
- Exponential signals, Decaying exponential signals ...
- Random waveforms

1. Overview of Time Domain Analysis Methods

- Classical Method for Solving Continuous-Time Systems
- Impulse response $h(t)$



- **Convolutional** method for **zero-state response** calculation

$$r_{zs}(t) = e(t) * h(t)$$

- Discrete-Time Systems: Analysis methods follow analogous principles

Main Content

1. Description of Linear Time-Invariant Systems
2. Solution Methods for System Differential Equations
3. Classical Method

Complete Solution = **Homogeneous** Solution + **Particular** Solution

2. System Differential Equations (for reference)

1. Description of Linear Time-Invariant Systems

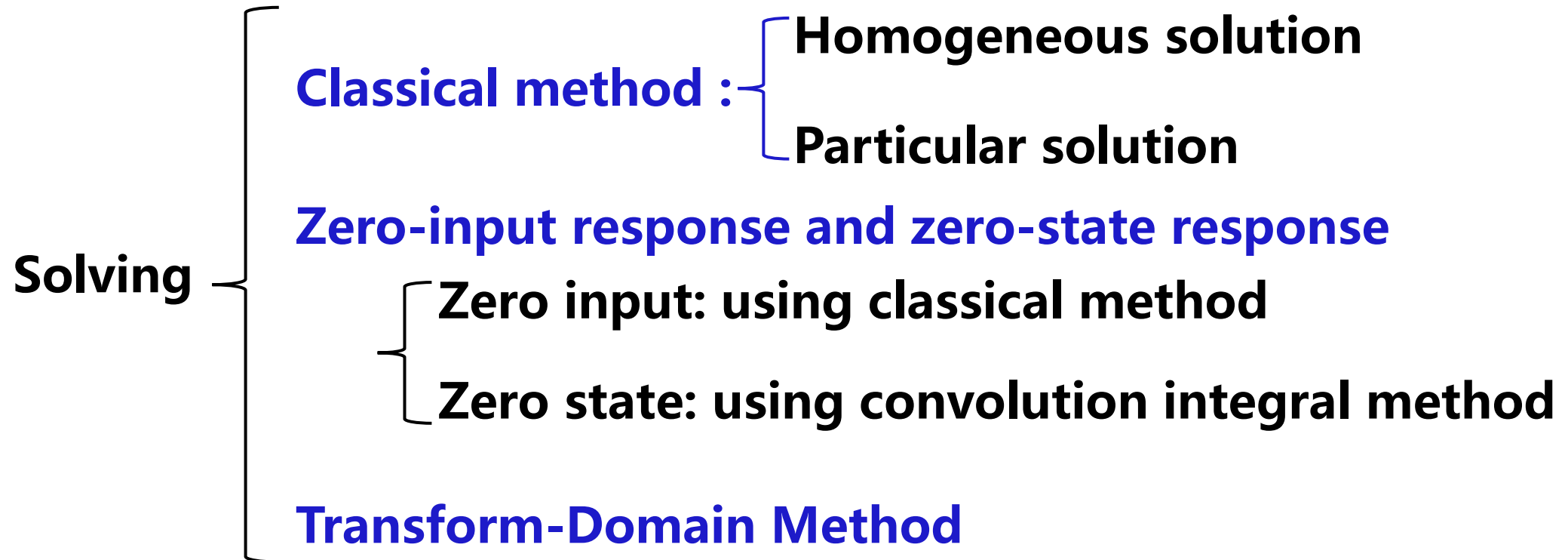
For a **LTI system**, the relationship between the excitation signal $e(t)$ and the response signal $r(t)$ can be described by the following **differential equation**

$$\begin{aligned} & a_n \frac{d^n r(t)}{dt^n} + a_{n-1} \frac{d^{n-1} r(t)}{dt^{n-1}} + \cdots + a_1 \frac{dr(t)}{dt} + a_0 r(t) \\ &= b_m \frac{d^m e(t)}{dt^m} + b_{m-1} \frac{d^{m-1} e(t)}{dt^{m-1}} + \cdots + b_1 \frac{de(t)}{dt} + b_0 e(t) \end{aligned}$$

n, m is an integer, and the coefficients are all constants

System Order: Determined by **the highest differential term**

2. Solution Methods for System Differential Equations



Summary

Classical method for solving differential equations:

- Complete solution = Homogeneous solution + Particular solution
- **Homogeneous solution method** :
Characteristic Equation $\rightarrow$ Characteristic Root $\rightarrow$ Homogeneous Solution
- **Particular solution method**
- The particular solution is in the same form as the input
- Undetermined coefficient: substituting it into the original equation
- **Determination of the coefficient of homogeneous solution**: using the Initial condition of the system

Main Content

1. Classification Methods for System Responses
2. Impulse Response and Step Response of Continuous-Time Systems

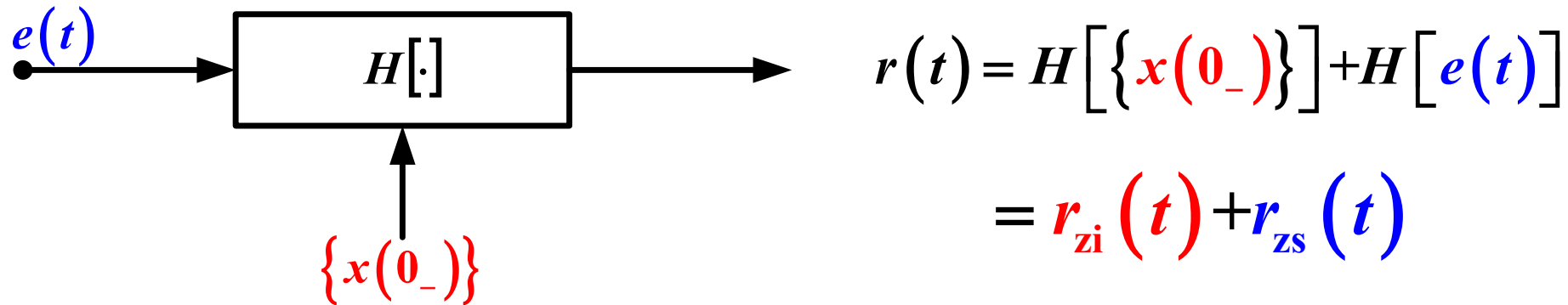
3. Continuous-Time System Response

1. Classification Methods for System Responses

- **Zero-input + Zero-state**
- **Natural + Forced**
- **Transient + Steady-state**

3. Continuous-Time System Response

1.1 Zero-input + Zero-state



(1) Zero-Input response (set input to zero)

Definition: Response produced exclusively by the system's initial energy storage (initial state)

Mathematics: Solution to the homogeneous differential equation with initial conditions

(2) Zero-State response (set the initial state to zero, no initial energy storage)

Definition: Response produced exclusively by the external excitation signal

Mathematics: Convolution of input with system's impulse response

(1) Solution of Zero-Input response

$$r_{zi}(t)$$

Input : 0

Equation : Homogeneous equation

Solution Form: Particular solution is zero, only homogeneous solution

Boundary conditions (without jumping) :

$$r_{zi}(0_-) = r(0_-), r_{zi}'(0_-) = r'(0_-), \dots, r_{zi}^{(n-1)}(0_-) = r^{(n-1)}(0_-)$$

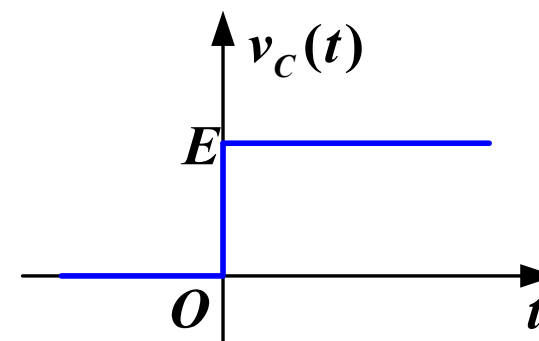
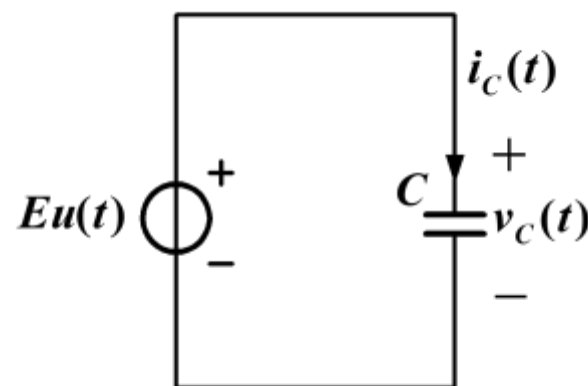
3. Continuous-Time System Response

(2) Jump Discontinuity at the Initial Point

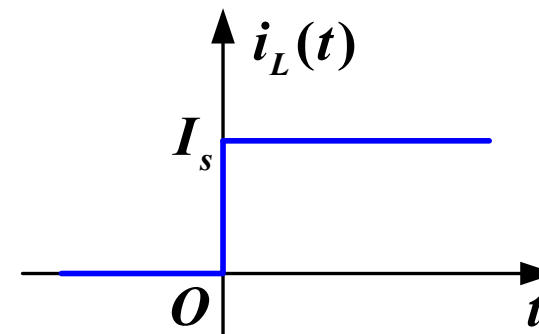
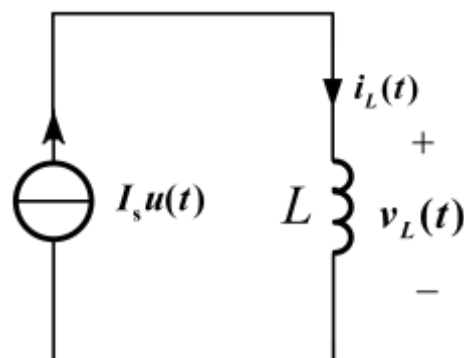
$$v_C(0_-) = 0$$

$$v_C(0_+) = E$$

$$i_C(t) = C \frac{dv_C(t)}{dt}$$
$$= CE\delta(t)$$



$$v_L(t) = L \frac{di_L(t)}{dt}$$
$$= LI_s\delta(t)$$



A sudden change in the system's state variables (e.g., capacitor voltage, inductor current) at the moment of excitation ($t = 0_+$), violating continuity conditions

3. Continuous-Time System Response

(3) Solution of Zero-state response

$$r_{zs}(t)$$

Zero State: $r_{zs}(0_-) = r'_{zs}(0_-) = \dots = r_{zs}^{(n-1)}(0_-) = 0$

Input : Given

Equation : Original

Solution Form: Particular solution + Homogeneous solution

Jumping: $r_{zs}(0_+), r'_{zs}(0_+), \dots, r_{zs}^{(n-1)}(0_+)$

Solution is complex, hence the convolution integral method is introduced

(4) Significance of Zero-State Response Analysis

- For most practical communication and electronic systems, only the **zero-state response** needs to be studied
- The zero-state response can be solved using the **convolution method**
- System "**identification**" and analysis of its inherent characteristics can be achieved through input-output relationships

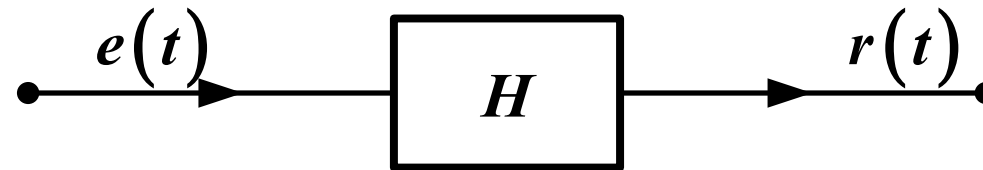
3. Continuous-Time System Response

2. Impulse Response and Step Response

- **Definition of Impulse response and Step response**
- **The impulse response of the typical systems**
- **Using Impulse response to judge the causality and stability of a system**

2.1 Definition of Impulse response and Step response

(1) Unit Impulse response



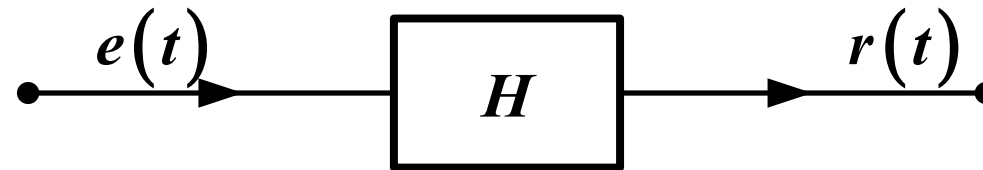
The zero-state response of a system excited by a unit impulse signal $\delta(t)$ is called the **unit impulse response** (or simply impulse response), denoted as $h(t)$

$$h^{(k)}(0_-) = 0 \quad (k = 0, 1, \dots, n-1)$$

$$\delta(t) \xrightarrow{\text{zs}} h(t)$$

2.1 Definition of Impulse response and Step response

(2) Unit step response



The zero-state response of a system excited by a unit step signal $u(t)$ is called the unit step response (or simply step response)

$$g^{(k)}(0_-) = 0 \quad (k = 0, 1, \dots, n-1)$$

$$u(t) \xrightarrow{\text{zs}} g(t)$$

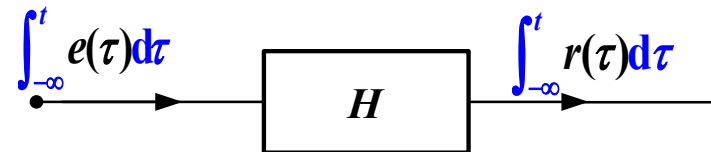
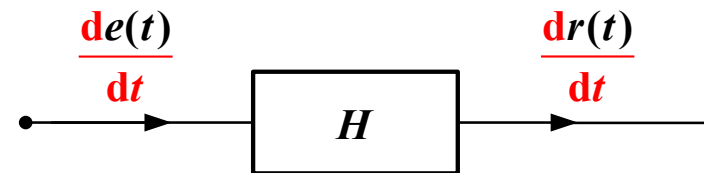
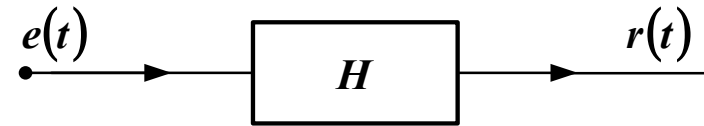
3. Continuous-Time System Response

2.1 Definition of Impulse response and Step response

(3) Impulse response vs Step response

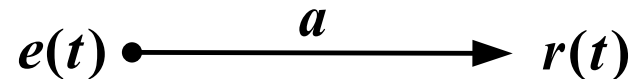
$$u(t) = \int_{-\infty}^t \delta(\tau) d\tau$$

$$g(t) = \int_{-\infty}^t h(\tau) d\tau$$



2.2 The impulse response of the typical systems

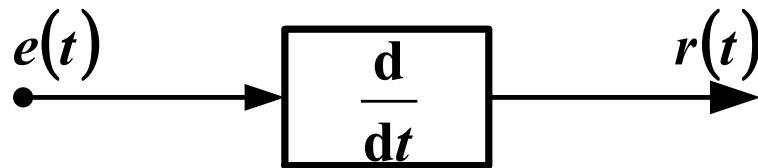
(1) Scalar multiplier (Digital multiplier, scaler)



$$r(t) = ae(t)$$

$$h(t) = a\delta(t)$$

(2) Differentiator

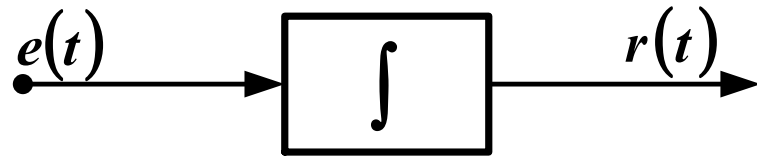


$$r(t) = \frac{de(t)}{dt}$$

$$h(t) = \frac{d\delta(t)}{dt} = \delta'(t)$$

2.2 The impulse response of the typical systems

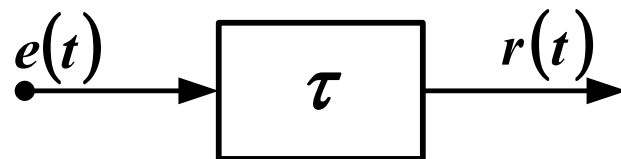
(3) Integrator



$$r(t) = \int_{-\infty}^t e(\tau) d\tau$$

$$h(t) = \int_{-\infty}^t \delta(\tau) d\tau \\ = u(t)$$

(4) Delayer (Unit Delay Element)



$$r(t) = e(t - \tau)$$

$$h(t) = \delta(t - \tau)$$

2.3 To judge the causality and stability of a system

The unit Impulse response of the causal system satisfies the following properties

$$h(t)=0 \quad (t < 0)$$

Applying the causality concept to signals, a signal is **causal**:

$$x(t) = 0 \quad (t < 0)$$

The unit impulse response of a **stable system** satisfies the following properties

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty$$

Summary

1. Classification Methods for System Responses

2. Impulse Response and Step Response of Continuous-Time Systems

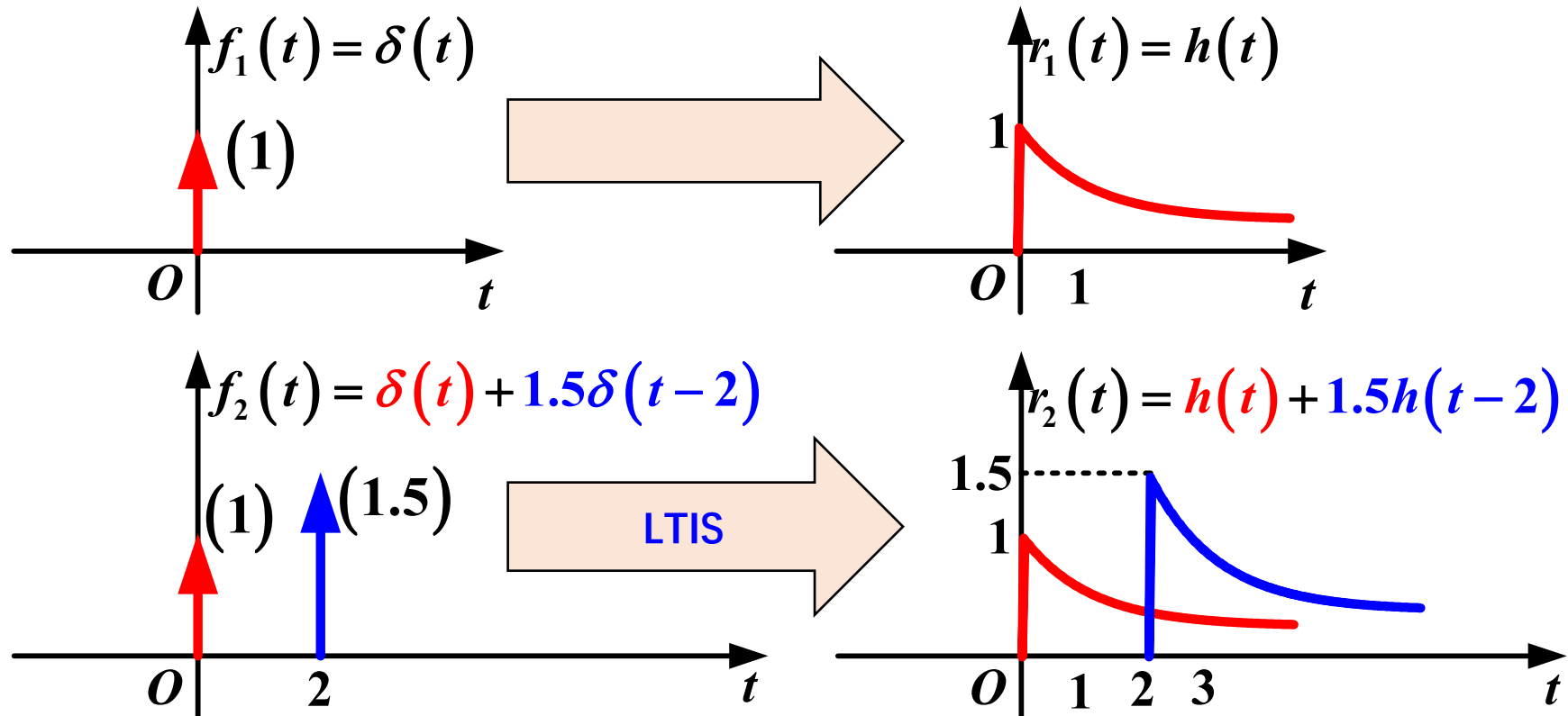
- Definition of Impulse response: **The zero-state response of a system under the action of a unit impulse signal $\delta(t)$ is called the impulse response**
- The impulse response can fully characterize system behavior
- Three time-domain solution methods for determining impulse response: (1) Singularity function balance method, (2) Time-Domain Direct Solution, (3) Convolution Integral
- The time-domain method is physically intuitive, but the calculation is cumbersome
- Calculation of impulse response and step response is simpler and more convenient using transform-domain methods (e.g., **Laplace transform**)

Main Content

1. Introduction to Convolution
2. Zero-State Response via Convolution
3. Physical Interpretation of Convolution
4. Properties of Convolution
5. Convolution Calculation

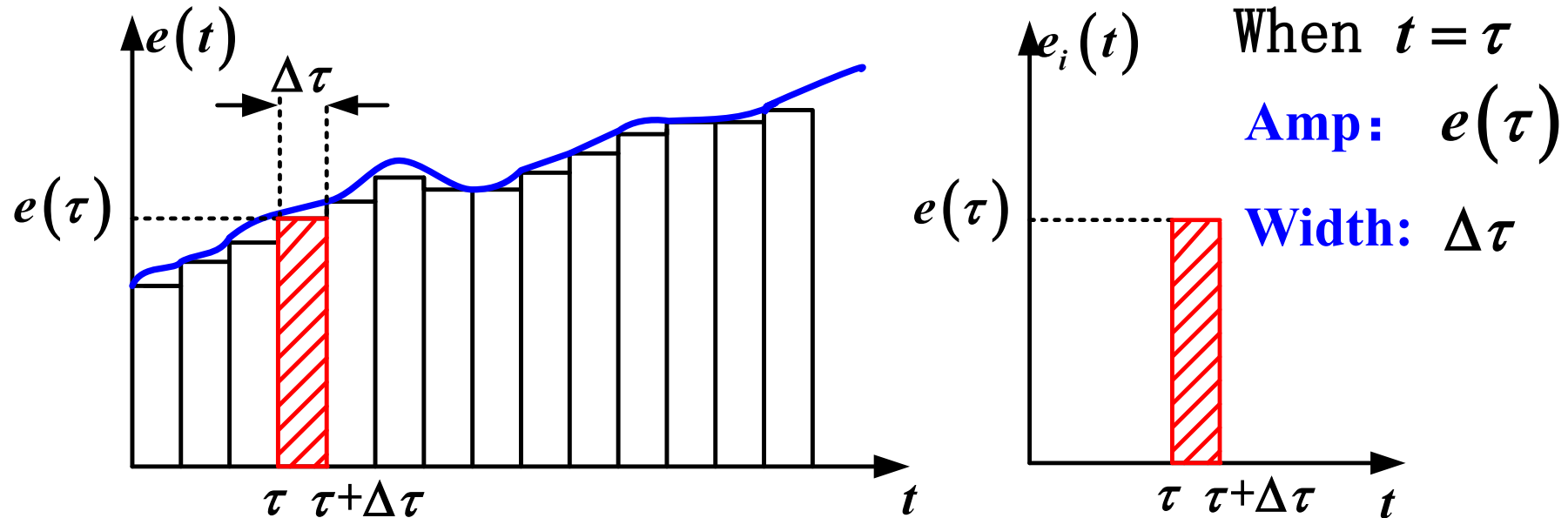
4. Continuous-Time Convolution

1. Introduction to Convolution



How to determine the **zero-state response** of a random signal $f(t)$ passing through the system?

1. Introduction to Convolution



Narrow pulse signals can be represented as

$$e_i(t) = e(\tau) [u(t - \tau) - u(t - \tau - \Delta\tau)]$$

From $\tau = -\infty$ to ∞ , $e(t)$ can be represented as the superposition of many narrow pulses

$$e(t) \approx \sum_{\tau=-\infty}^{\infty} e(\tau) [u(t - \tau) - u(t - \tau - \Delta\tau)]$$

4. Continuous-Time Convolution

1. Introduction to Convolution

$$e(t) \approx \sum_{\tau=-\infty}^{\infty} e(\tau) \frac{[u(t-\tau) - u(t-\tau-\Delta\tau)]}{\Delta\tau} \cdot \Delta\tau$$

Let $\Delta\tau \rightarrow 0$

$$\Delta\tau \rightarrow d\tau$$

$$\sum_{\tau=-\infty}^{\infty} \rightarrow \int_{\tau=-\infty}^{\infty}$$

$$\lim_{\Delta\tau \rightarrow 0} \frac{[u(t-\tau) - u(t-\tau-\Delta\tau)]}{\Delta\tau} = \frac{du(t-\tau)}{dt} = \delta(t-\tau)$$

difference

differential

Represent **any** signal as a **weighted superposition** of **time-shifted unit impulse signals**

$$e(t) = \int_{-\infty}^{\infty} e(\tau) \delta(t-\tau) d\tau$$

2. Zero-State Response via Convolution

A mathematical operation that combines two functions (typically a signal and a system's impulse response) to produce a third function representing how one modifies the other

A signal $e(t)$ can be represented as the sum of impulse sequences

$$e(t) = \int_{-\infty}^{\infty} e(\tau) \delta(t - \tau) d\tau$$

If it is applied to LTIS with an impulse response of $h(t)$, the response is

$$\delta(t) \rightarrow h(t)$$

$$\delta(t - \tau) \rightarrow h(t - \tau)$$

Time-Invariance

Linearity

$$e(\tau) \delta(t - \tau) \rightarrow e(\tau) h(t - \tau)$$

Zero-state
response

$$e(t) = \int_{-\infty}^{\infty} e(\tau) \delta(t - \tau) d\tau \rightarrow \int_{-\infty}^{\infty} e(\tau) h(t - \tau) d\tau = r(t)$$

2. Zero-State Response via Convolution

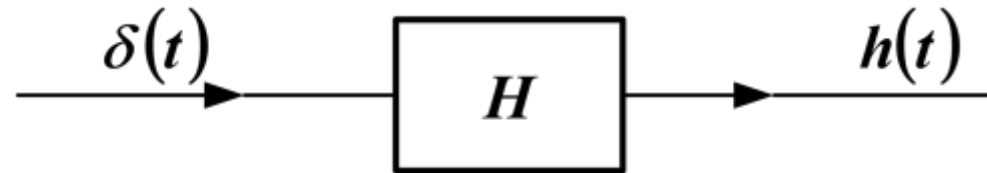
With two functions $f_1(t)$ and $f_2(t)$, integral

$$f(t) = \int_{-\infty}^{\infty} f_1(\tau) f_2(t - \tau) d\tau$$

Convolution integral of $f_1(t)$ and $f_2(t)$, abbreviated as **convolution**, denoted as

$$f(t) = f_1(t) \otimes f_2(t) \quad \text{or} \quad f(t) = f_1(t) * f_2(t)$$

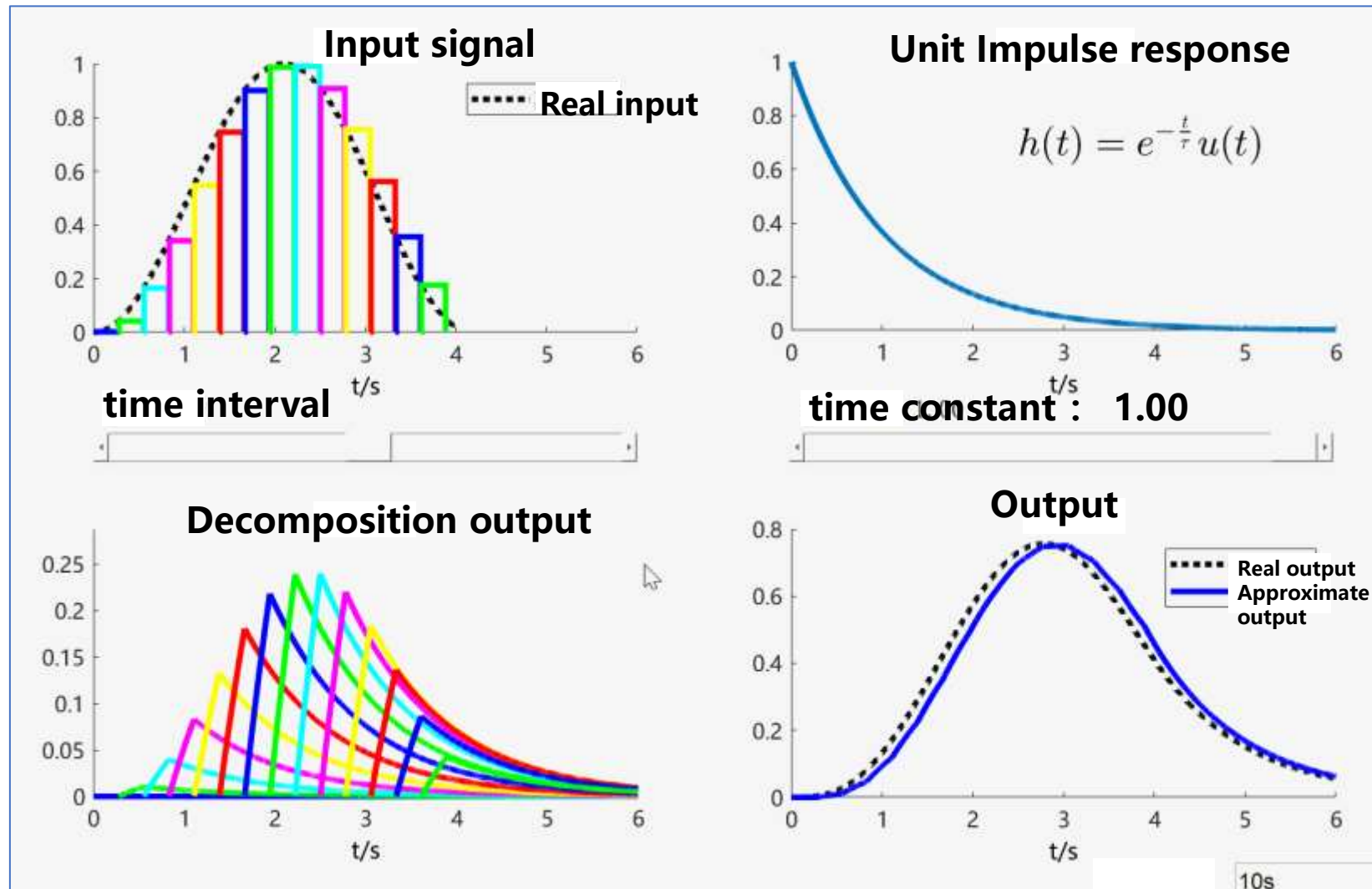
The zero-state response of the system can be solved using convolution:



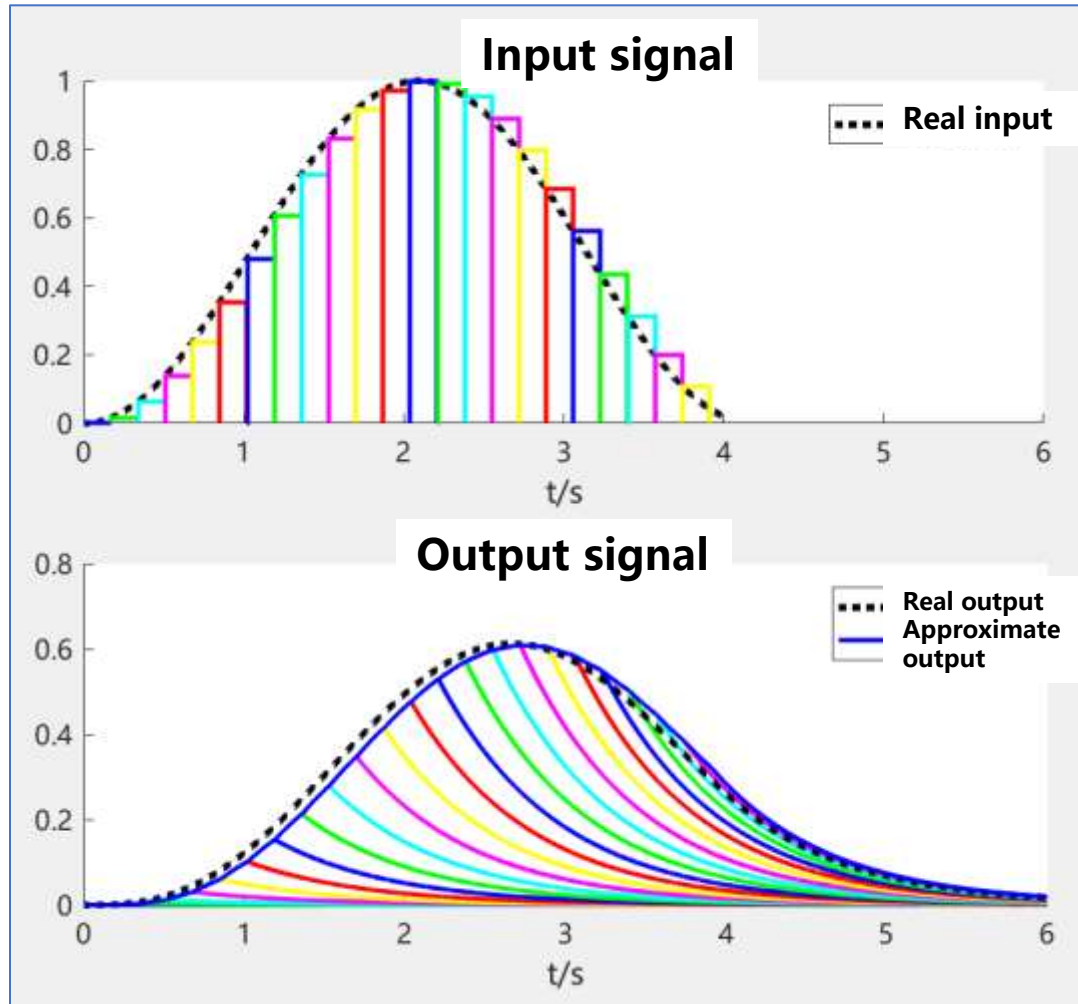
$$r_{zs}(t) = \int_{-\infty}^{\infty} e(\tau) h(t - \tau) d\tau$$

4. Continuous-Time Convolution

3. Physical Interpretation of Convolution



3. Physical Interpretation of Convolution



$$r_{zs}(t) = \int_{-\infty}^{\infty} e(\tau) h(t - \tau) d\tau$$

τ : The moment when the signal occurs

$e(\tau)$: excitation at time t

t : Time to observe the response

$h(t - \tau)$: The system's impulse response at time t due to a unit impulse applied at τ

4. Properties of Convolution

- **Algebraic properties of convolution**
 - (1) **Commutative Property**
 - (2) **Distributive Property and Parallel Systems**
 - (3) **Associative Property and Series Systems**
- **Time-Shift Property**
- **Convolution with Impulse/Step Functions**
- **Differentiation & Integration Properties**

4.1 Algebraic properties of convolution

(1) Commutative property

$$f_1(t) * f_2(t) = f_2(t) * f_1(t)$$

proof:

$$f_1(t) * f_2(t) = \int_{-\infty}^{\infty} f_1(\tau) \cdot f_2(t - \tau) d\tau$$

let $t - \tau = \lambda$

Thus $\tau = t - \lambda$

$$d\tau = -d\lambda$$

$$\tau : \int_{-\infty}^{\infty} \rightarrow \lambda : \int_{\infty}^{-\infty}$$

$$= \int_{\infty}^{-\infty} f_2(\lambda) \cdot f_1(t - \lambda) (-d\lambda)$$

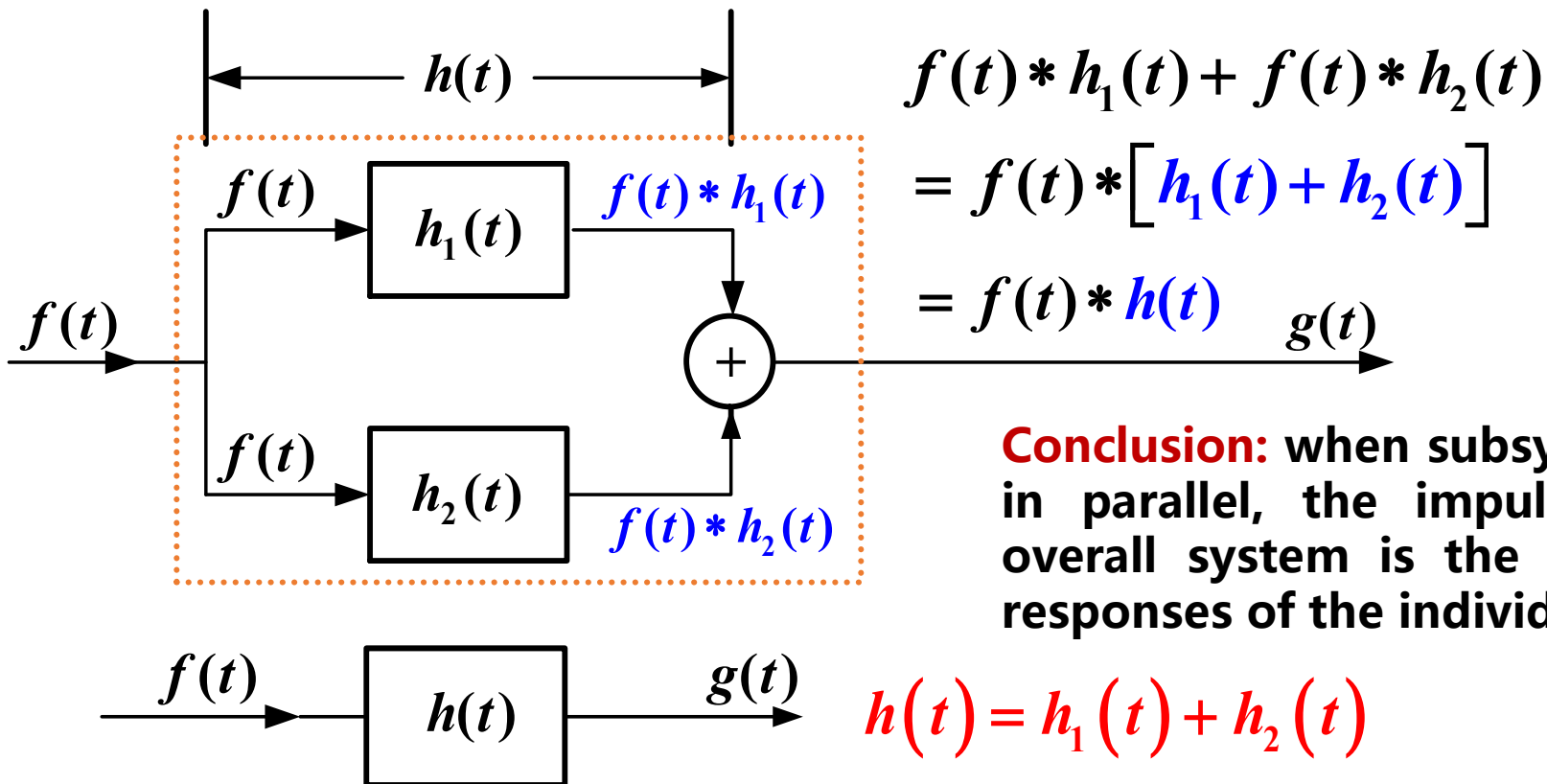
$$= \int_{-\infty}^{\infty} f_2(\lambda) \cdot f_1(t - \lambda) d\lambda$$

$$= f_2(t) * f_1(t)$$

4.1 Algebraic properties of convolution

(2) Distributive Property and Parallel Systems

$$f_1(t) * [f_2(t) + f_3(t)] = f_1(t) * f_2(t) + f_1(t) * f_3(t)$$



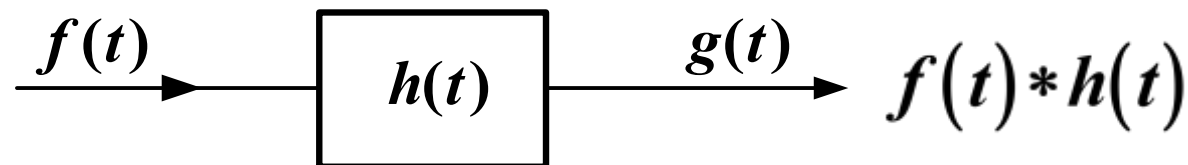
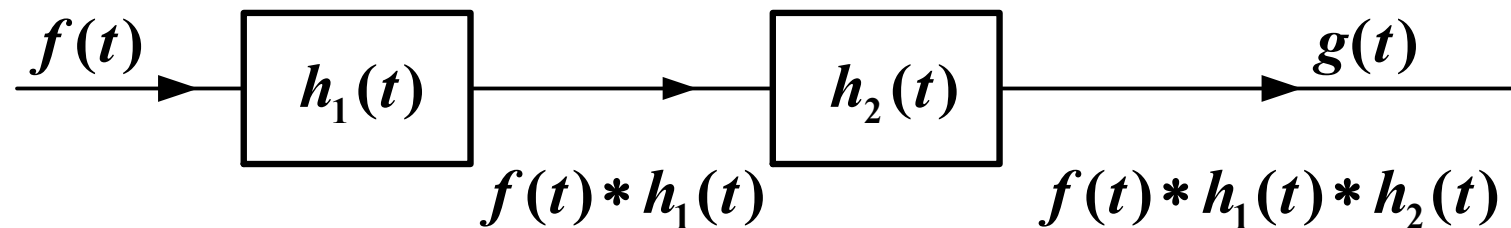
Conclusion: when subsystems are connected in parallel, the impulse response of the overall system is the sum of the impulse responses of the individual subsystems

4.1 Algebraic properties of convolution

(3) Associative Property and Series Systems

$$f(t) * h_1(t) * h_2(t) = f(t) * [h_1(t) * h_2(t)]$$

The system is connected in series, and the block diagram shows:



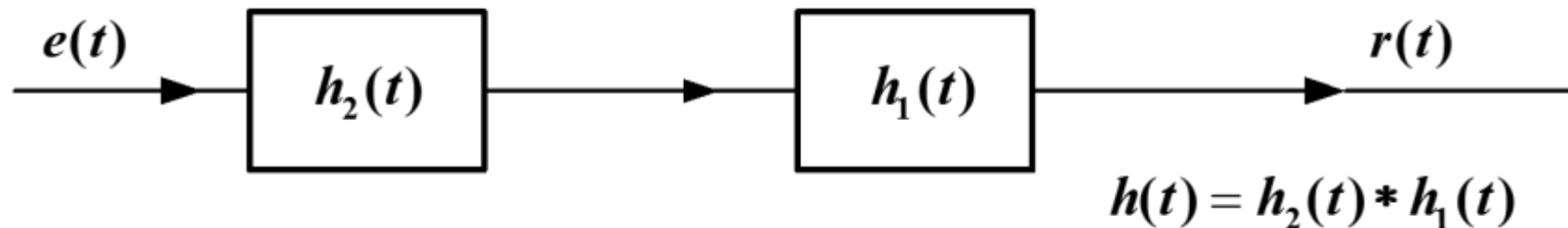
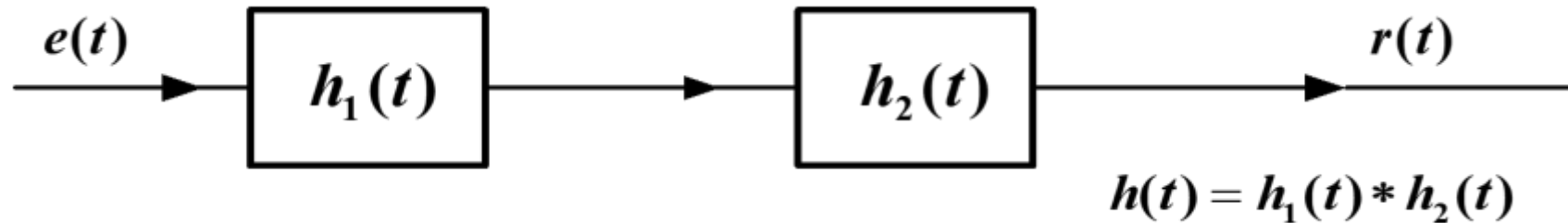
$$h(t) = h_1(t) * h_2(t)$$

Conclusion: when subsystems are connected in series (cascade), the impulse response of the overall system is the convolution of the individual impulse responses

4. Continuous-Time Convolution

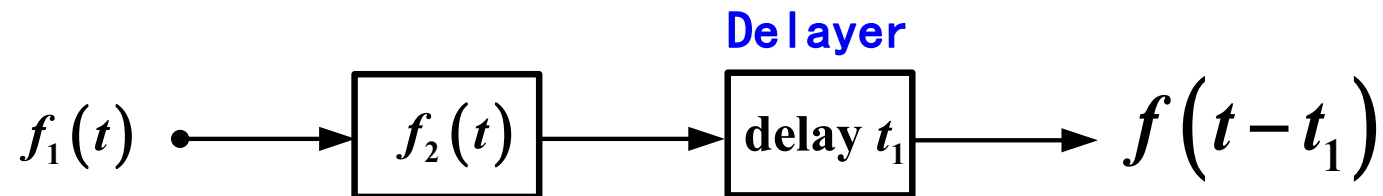
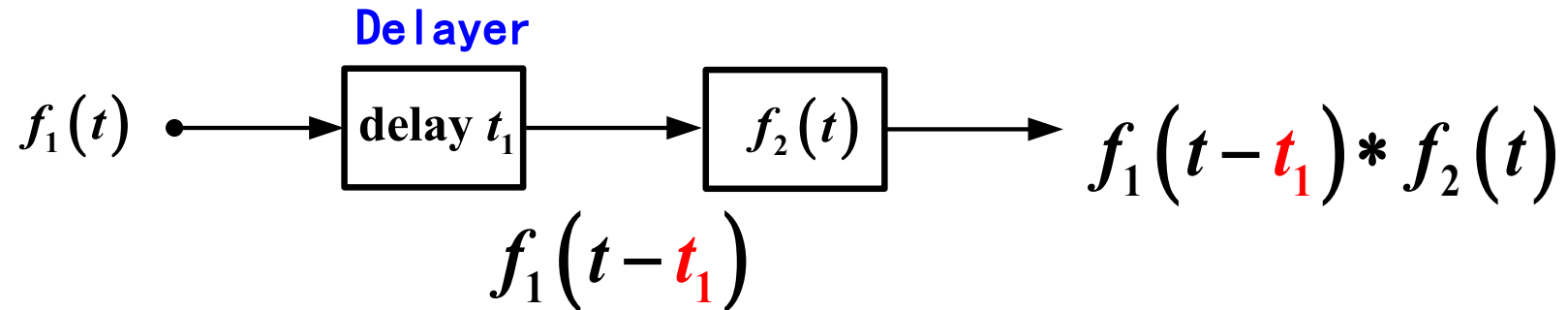
4.1 Algebraic properties of convolution

The order of subsystems connected in series (cascade) can be swapped



4.2 Time-Shift Property

If $f(t) = f_1(t) * f_2(t)$, Then $f_1(t - \mathbf{t}_1) * f_2(t) = f(t - \mathbf{t}_1)$



$$f(t) = f_1(t) * f_2(t)$$

$$f_1(t - \mathbf{t}_1) * f_2(t - \mathbf{t}_2) = f(t - \mathbf{t}_1 - \mathbf{t}_2)$$

4.3 Convolution with Impulse/Step Functions

a. $f(t) * \delta(t) = \int_{-\infty}^{\infty} f(t - \tau) \delta(\tau) d\tau$

$$= f(t) \int_{-\infty}^{\infty} \delta(\tau) d\tau = f(t)$$

$$f(t) \xrightarrow[h(t) = \delta(t)]{1} f(t)$$

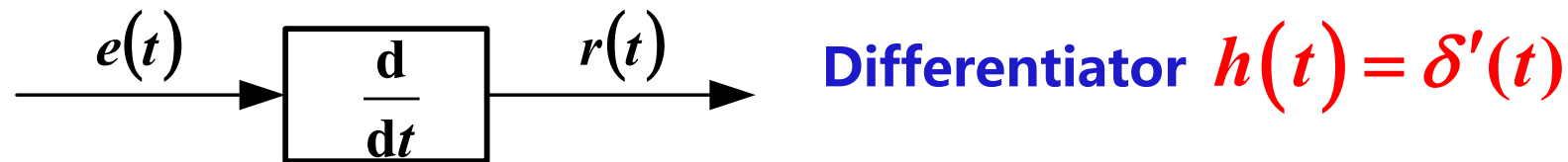
The unit Impulse response of an ideal conductor is $\delta(t)$

b. $f(t) * \delta(t - t_0) = f(t - t_0)$



4.3 Convolution with Impulse/Step Functions

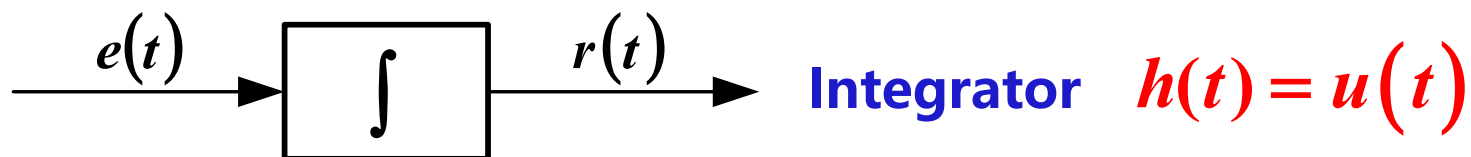
c. $f(t) * \delta'(t) = f'(t)$



$f(t) * \delta^{(k)}(t) = f^{(k)}(t)$ **K numbers differentiators**

$f(t) * \delta^{(k)}(t - t_0) = f^{(k)}(t - t_0)$ **Differential and Time Shift**

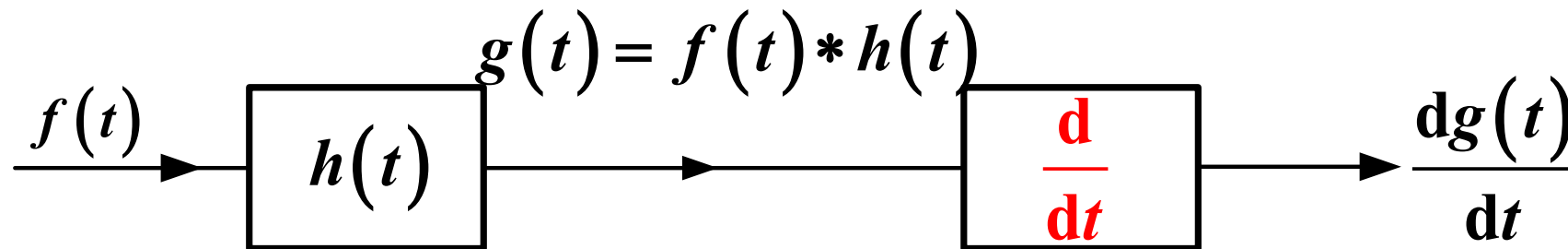
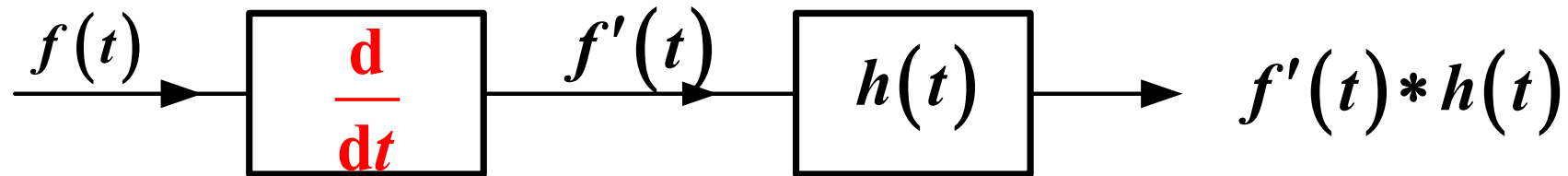
d. $f(t) * u(t) = \int_{-\infty}^t f(\lambda) d\lambda$



4.4 Differentiation & Integration Properties

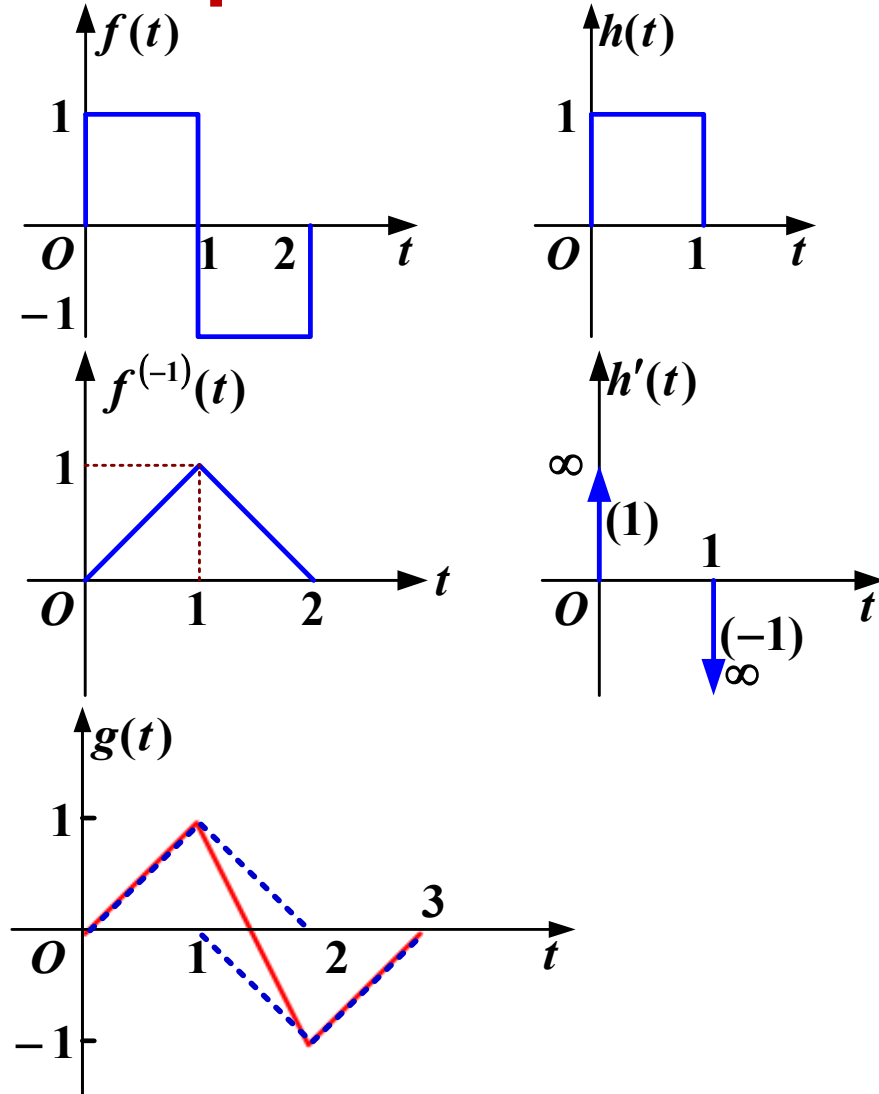
If $g(t) = f(t) * h(t)$

Then $g'(t) = f'(t) * h(t) = f(t) * h'(t)$



4. Continuous-Time Convolution

Example 1



If $f(t), h(t)$, get $g(t) = f(t) * h(t)$

$$g(t) = f^{(-1)}(t) * h'(t)$$

$$= f^{(-1)}(t) * [\delta(t) - \delta(t-1)]$$

$$= f^{(-1)}(t) - f^{(-1)}(t-1)$$

Convolution with $\delta(t)$: a weighted and shifted version of the original signal

4. Continuous-Time Convolution

5. Convolution Calculation-Analytical Method

$$r_{zs}(t) = \int_{-\infty}^{\infty} e(\tau) h(t - \tau) d\tau$$

Due to the causality of the system or the time-limited nature of the excitation signal, the limits of integration in convolution may vary. Determining the **correct integration limits** in convolution integrals is crucial

4. Continuous-Time Convolution

Example 2

Given $e(t) = u(t) - u(t - t_0)$,

$h(t) = e^{-t} u(t)$,

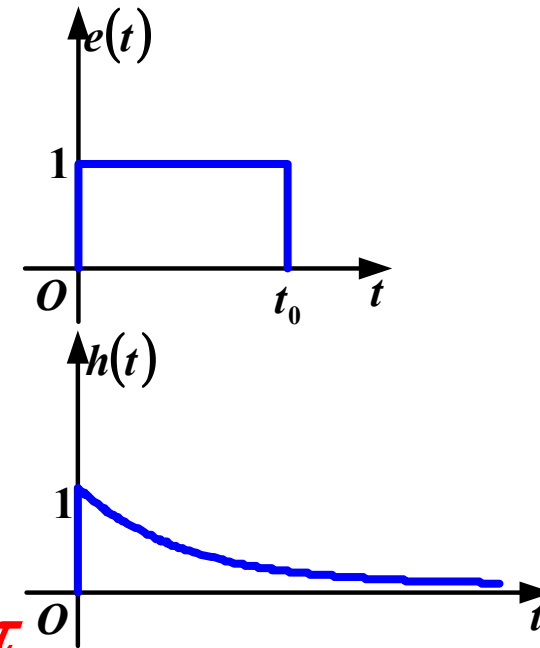
Solve $r(t) = e(t) * h(t)$, Let $t_0 > 0$.

$$r(t) = \int_{-\infty}^{\infty} e(\tau) \cdot h(t - \tau) d\tau$$

$$= \int_{-\infty}^{\infty} [u(\tau) - u(\tau - t_0)] \cdot e^{-(t-\tau)} u(t - \tau) d\tau$$

$$= e^{-t} \int_{-\infty}^{\infty} e^{\tau} [u(\tau) u(t - \tau)] d\tau - e^{-t} \int_{-\infty}^{\infty} e^{\tau} [u(\tau - t_0) u(t - \tau)] d\tau$$

$$= r_1(t) - r_2(t)$$



4. Continuous-Time Convolution

Example 2 Determining Integration Limits (Crucial)

$$r_1(t) = e^{-t} \int_{-\infty}^{\infty} e^{\tau} [u(\tau)u(t-\tau)] d\tau$$

$$= \left[e^{-t} \int_0^t e^{\tau} d\tau \right] \cdot u(t)$$

$$= (1 - e^{-t}) u(t)$$

$$u(\tau) \cdot u(t - \tau) \neq 0$$

$$\Rightarrow \begin{cases} \tau > 0 \\ t - \tau > 0 \end{cases}$$

$$\Rightarrow \begin{cases} 0 < \tau < t \\ t > 0 \end{cases}$$

4. Continuous-Time Convolution

Example 2

$$r_2(t) = e^{-t} \int_{-\infty}^{\infty} e^{\tau} [u(\tau - t_0)u(t - \tau)] d\tau$$

$$= \left[e^{-t} \int_{t_0}^t e^{\tau} d\tau \right] \cdot u(t - t_0)$$

$$= [1 - e^{-(t-t_0)}] u(t - t_0)$$

$$u(\tau - t_0) \cdot u(t - \tau) \neq 0$$

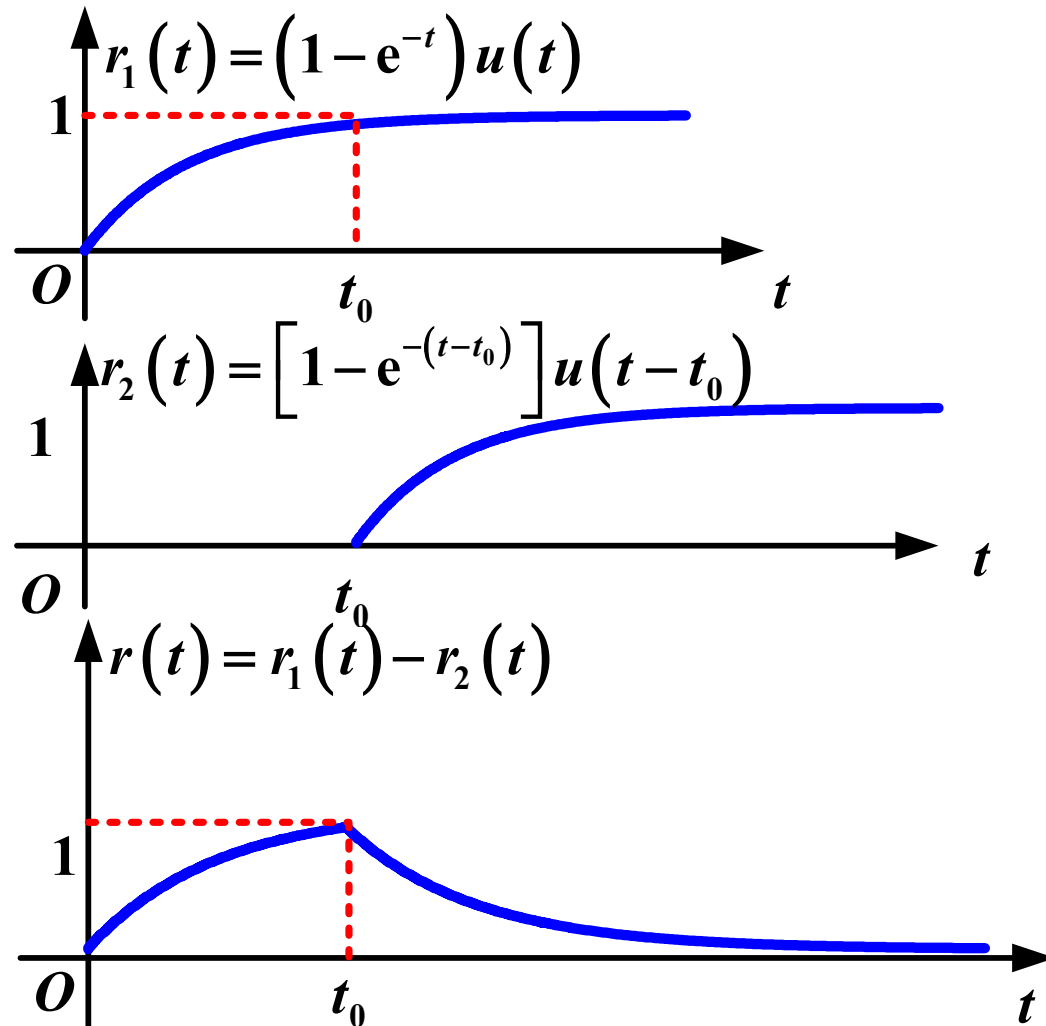
$$\Rightarrow \begin{cases} \tau - t_0 > 0 \\ t - \tau > 0 \end{cases}$$

$$\Rightarrow \begin{cases} t_0 < \tau < t \\ t > t_0 \end{cases}$$

4. Continuous-Time Convolution

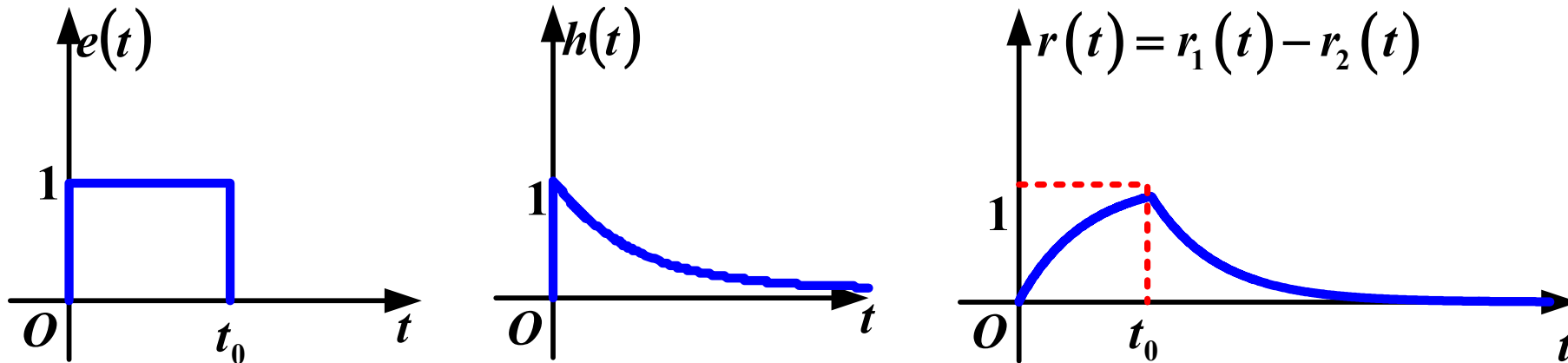
Example 2

$$\begin{aligned} r(t) &= r_1(t) - r_2(t) \\ &= (1 - e^{-t})u(t) \\ &\quad - [1 - e^{-(t-t_0)}]u(t-t_0) \end{aligned}$$



4. Continuous-Time Convolution

Example 2



- The output exhibits a tendency to "catch up" with the input
- This is a system with memory, where the output depends not only on the current input but also on past inputs

4. Continuous-Time Convolution

5. Convolution Calculation-Graphical Method

When the functional formula is complex, using the **graphical method** makes it easier to intuitively determine the **integration limits**

$$g(t) = \int_{-\infty}^{\infty} f_1(\tau) f_2(t - \tau) d\tau$$

1. $f_1(t) \rightarrow f_1(\tau)$, **change the integral variable to τ**

2. $f_2(t) \rightarrow f_2(\tau) \xrightarrow{\text{reversal}} f_2(-\tau) \xrightarrow{\text{time delay}} f_2(t - \tau)$

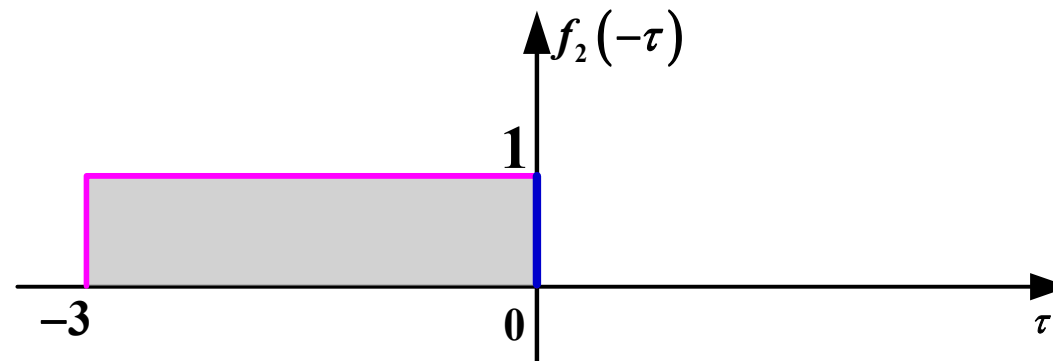
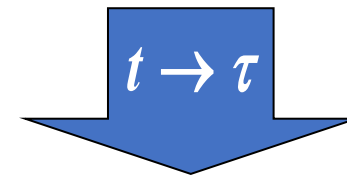
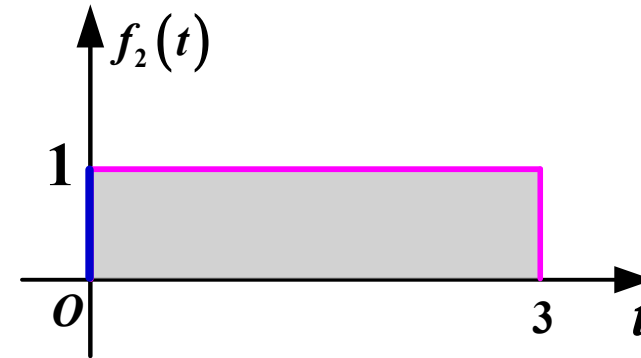
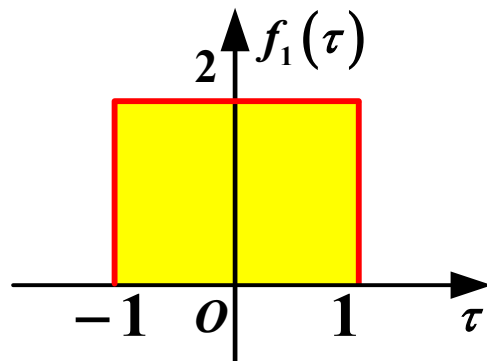
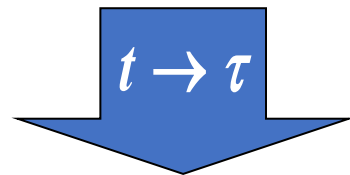
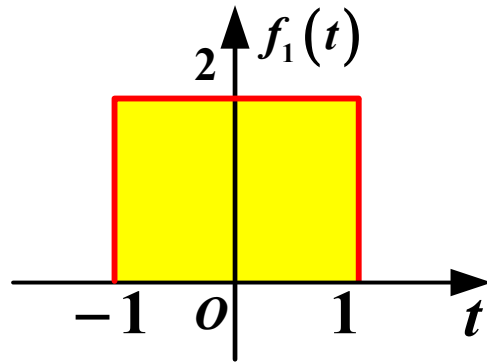
3. **Multiplication** $f_1(\tau) \cdot f_2(t - \tau)$

4. **Integral of product** $\int_{-\infty}^{\infty} f_1(\tau) \cdot f_2(t - \tau) d\tau$

4. Continuous-Time Convolution

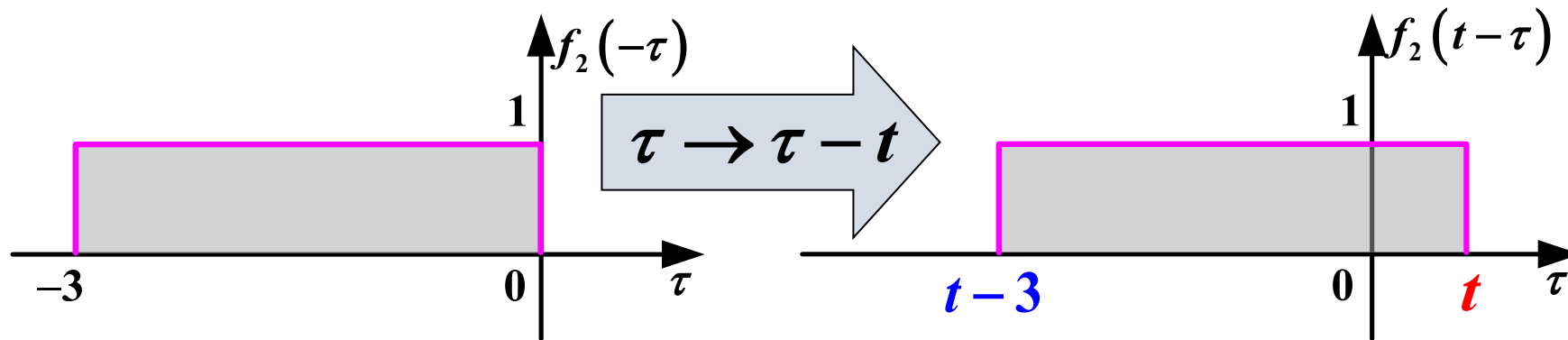
Example 3

The waveforms of $f_1(t)$ and $f_2(t)$ are as follows, find the convolution of $f_1(t)$ and $f_2(t)$, $g(t)$



4. Continuous-Time Convolution

Example 3



$$f_2(t - \tau) = f_2[-(\tau - t)]$$

Original coordinate system New coordinate system

$$\tau = 0$$

$$\tau = t$$

$$\tau = -3$$

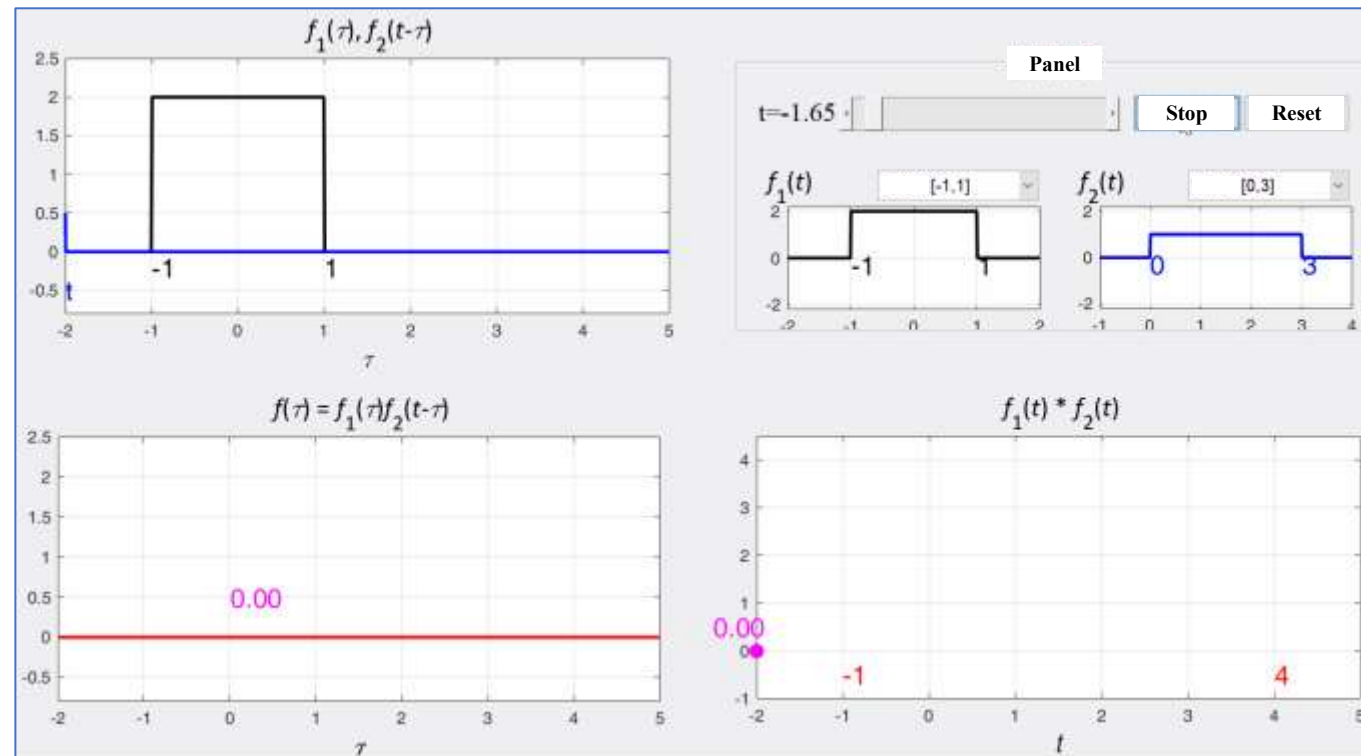
$$\tau = t - 3$$

4. Continuous-Time Convolution

Example 3

$$g(t) = \int_{-\infty}^{\infty} f_1(\tau) f_2(t - \tau) d\tau$$

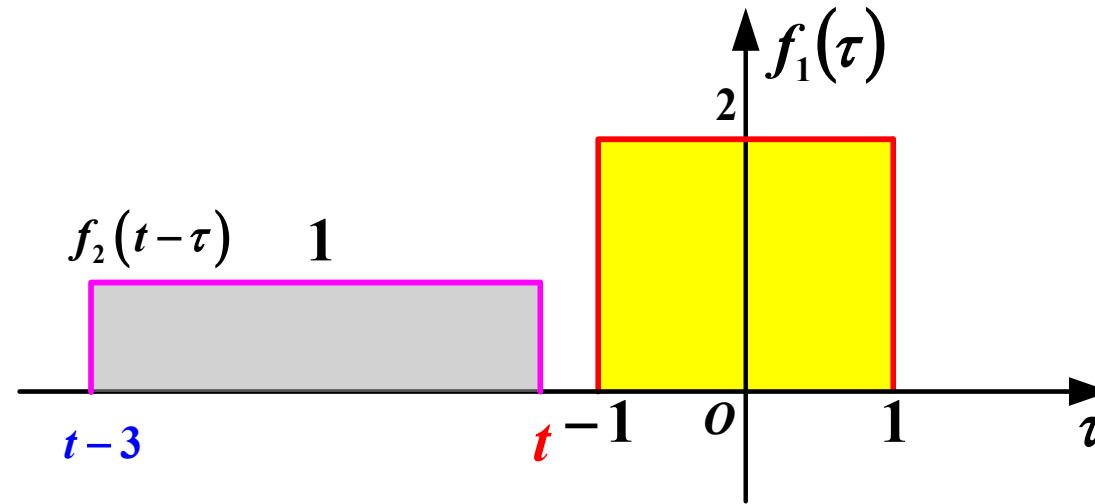
t from $-\infty$ to $+\infty$ corresponds to $f(t-\tau)$ move from left to right



**Multiply and
then integrate**

4. Continuous-Time Convolution

Example 3



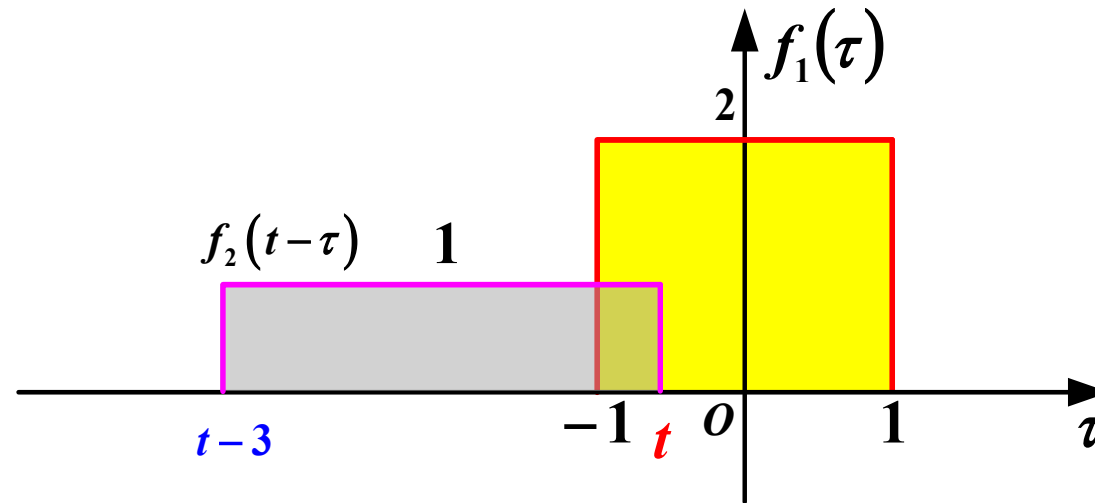
① $t \leq -1$

$$f_1(\tau) \cdot f_2(t-\tau) = 0$$

$$g(t) = f_1(t) * f_2(t) = 0$$

4. Continuous-Time Convolution

Example 3

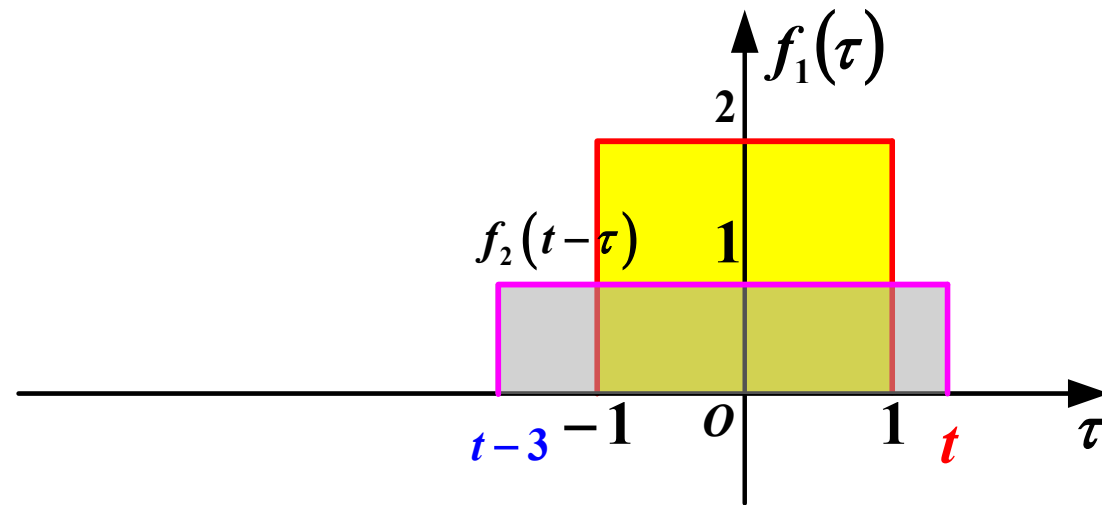


$$\textcircled{2} \begin{cases} t \geq -1 \\ t \leq 1 \end{cases} \Rightarrow -1 \leq t \leq 1$$

$$\begin{aligned} g(t) &= \int_{-1}^t f_1(\tau) \cdot f_2(t-\tau) d\tau \\ &= \int_{-1}^t 2 d\tau = 2t \Big|_{-1}^t = 2t + 2 \end{aligned}$$

4. Continuous-Time Convolution

Example 3

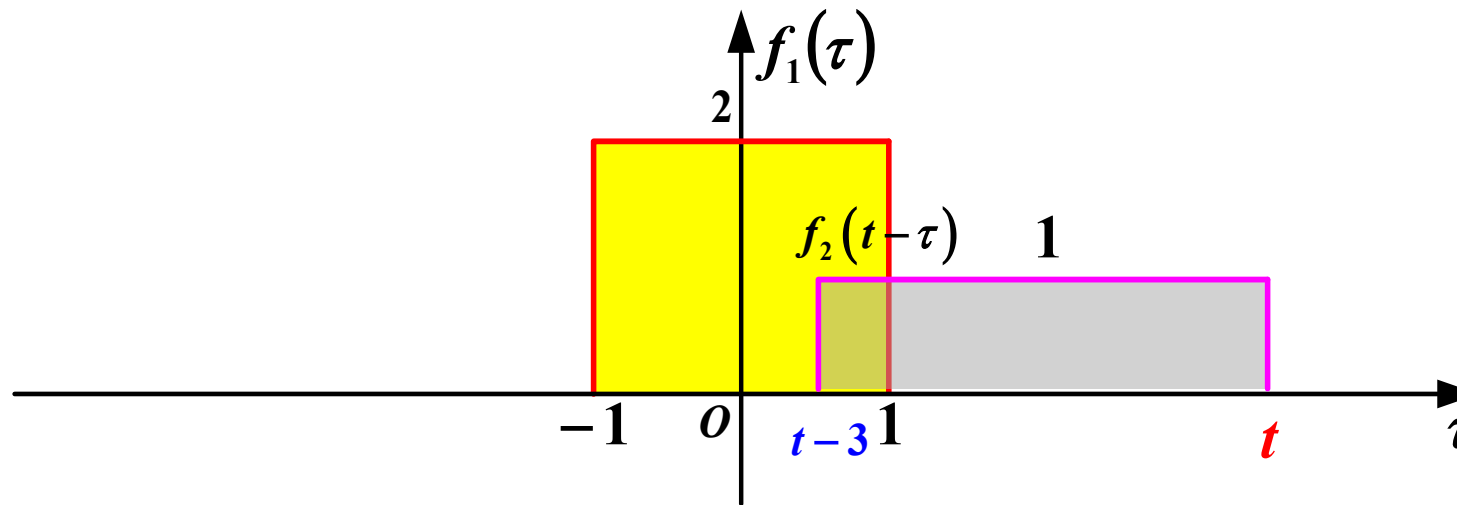


③ $\begin{cases} t \geq 1 \\ t-3 \leq -1 \end{cases}$ Thus $1 \leq t \leq 2$

$$\begin{aligned} g(t) &= \int_{-1}^1 2 d\tau \\ &= 4 \end{aligned}$$

4. Continuous-Time Convolution

Example 3



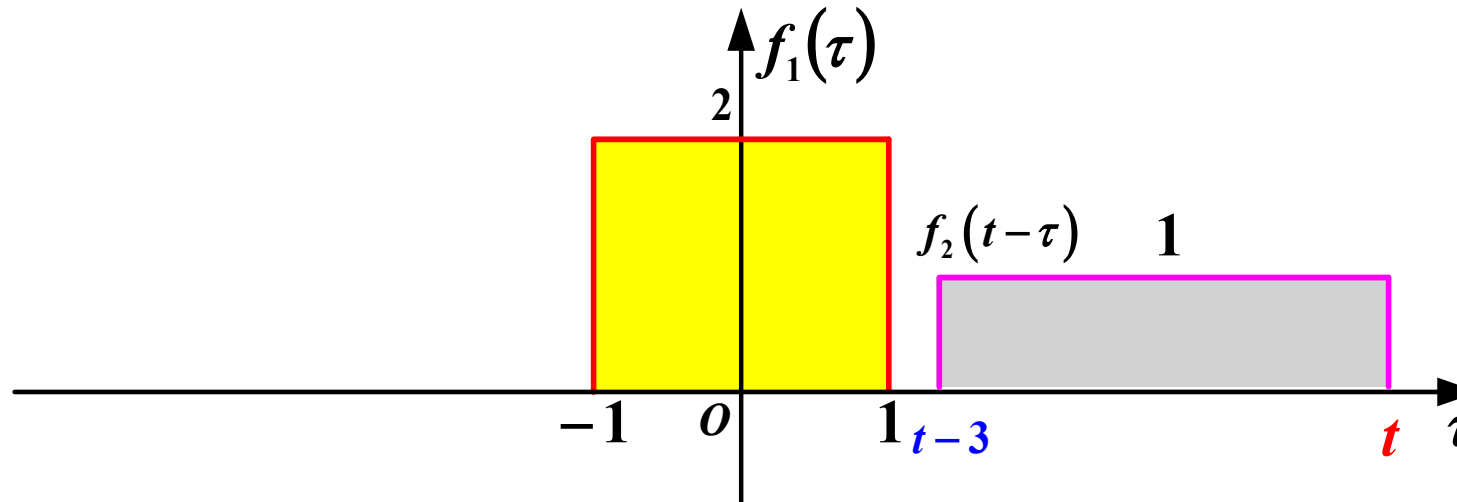
$$\textcircled{4} \begin{cases} t-3 \geq -1 \\ t-3 \leq 1 \end{cases} \quad \text{Thus } 2 \leq t \leq 4$$

$$g(t) = \int_{t-3}^1 2 d\tau = 2\tau \Big|_{t-3}^1$$

$$= -2t + 8$$

4. Continuous-Time Convolution

Example 3

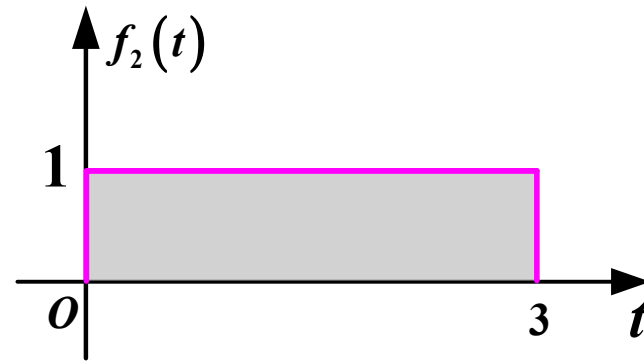
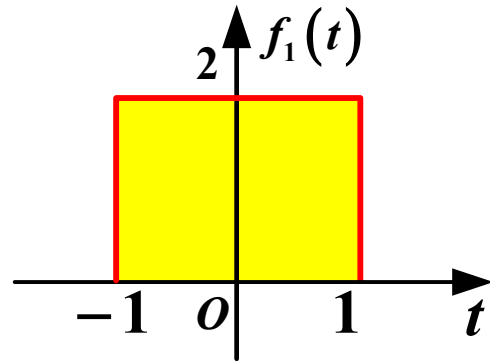


⑤ $t - 3 \geq 1$ Thus $t \geq 4$

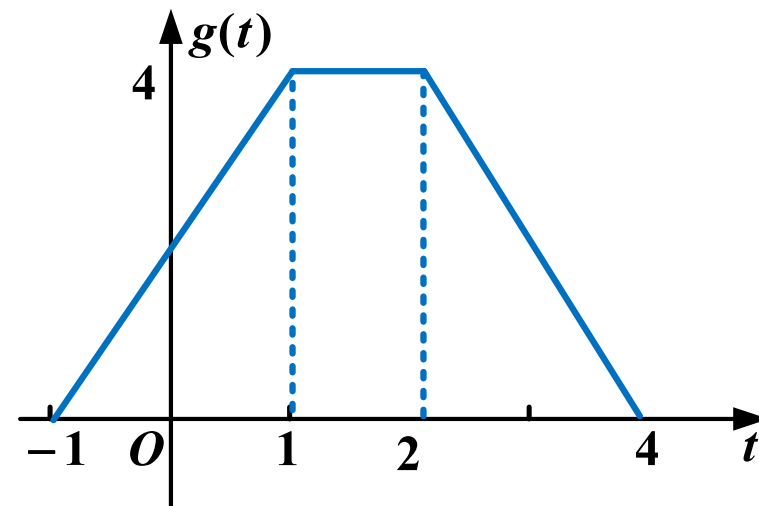
$$g(t) = 0$$

4. Continuous-Time Convolution

Example 3

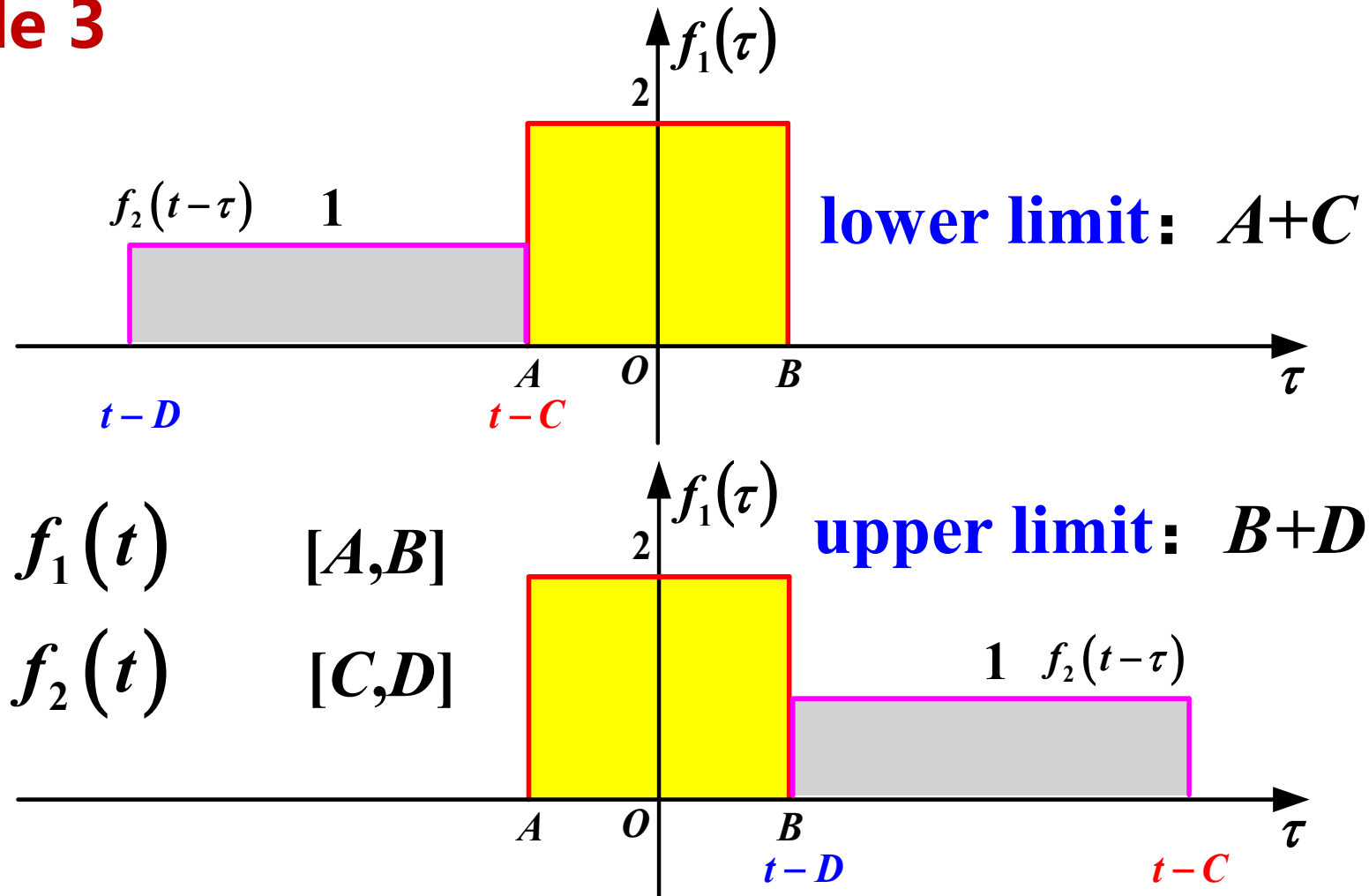


$$g(t) = \begin{cases} \textcircled{2} = 2t + 2 & -1 \leq t \leq 1 \\ \textcircled{3} = 4 & 1 \leq t \leq 2 \\ \textcircled{4} = -2t + 8 & 2 \leq t \leq 4 \\ \textcircled{1} \textcircled{5} = 0 & \text{else } t \end{cases}$$



4. Continuous-Time Convolution

Example 3



The width of $g(t)$ = the sum of the widths of the two signals

Summary

1. Introduction to Convolution
2. Zero-State Response via Convolution
3. Physical Interpretation of Convolution
4. Properties of Convolution
5. Convolution Calculation

$$\begin{aligned}r(t) &= e(t) \otimes h(t) \\&= e(t) * h(t) \\&= \int_{-\infty}^{\infty} e(\tau) h(t - \tau) d\tau\end{aligned}$$

Main Content

- **Iterative method**
- **Classical Time-Domain Method : Homogeneous solution + Particular solution (Similar to Continuous Systems)**
- **Zero-Input response + Zero-State response (Similar to Continuous Systems)**
Using convolutional sum to calculate the zero-state response of a system
- **Z-transform method $\rightarrow$ Inverse transform $\rightarrow y(n)$ (Chapter 6)**

1. Iterative method

- The difference equation itself is a recursive relationship
- But cannot obtain an analytical expression for the output sequence
- Suitable for using computers to solve problems

5. Difference Equations

Example 1

Given $y[n] = 0.9y[n-1] + u[n]$, $y[-1] = 0$, solve the equation

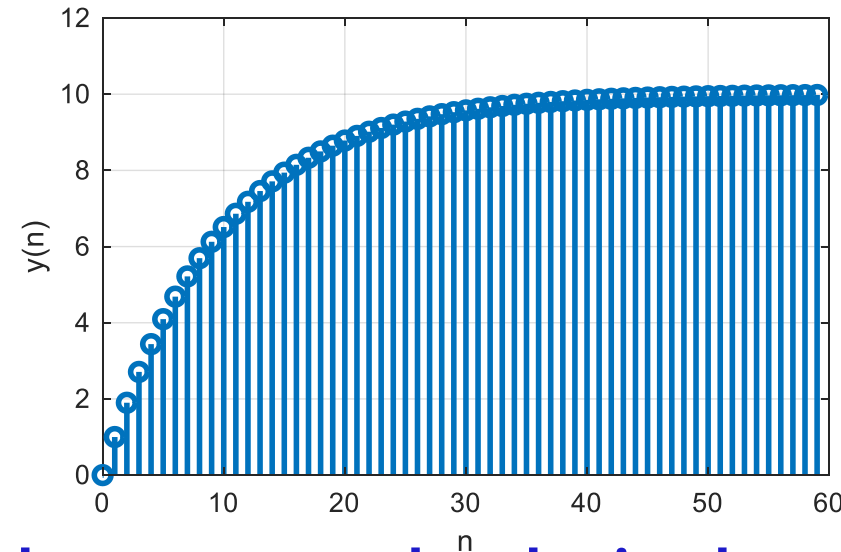
$$n=0 \quad y[0] = 0.9y[-1] + 1 = 1$$

$$n=1 \quad y[1] = 0.9y[0] + 1 = 1.9$$

$$n=2 \quad y[2] = 0.9y[1] + 1 = 2.71$$

$$n=3 \quad y[3] = 0.9y[2] + 1 = 3.439$$

...



From the recursive relationship, the output can be obtained

$$y[n] = \left\{ \underset{\substack{\uparrow \\ n=0}}{1}, 1.9, 2.71, 3.439, \dots \right\}$$

Main Content

- Definition and Solution of the Unit Impulse Response for Discrete Time Systems
- Judging Causality and Stability from $h[n]$

1. Definition and Solution of the Unit Impulse Response

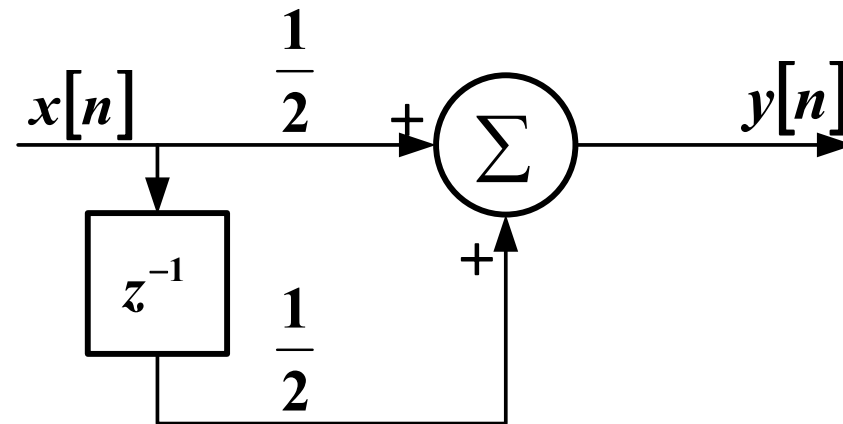


- Input $\delta[n]$
- **Zero-State:** $h[-n] = 0 \quad n = 1, 2, 3, \dots, N$

6. Unit Impulse Response

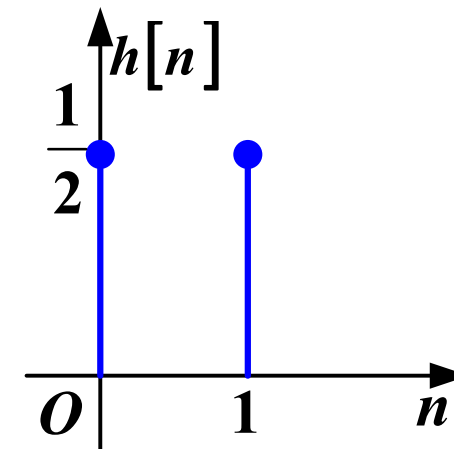
Example 1

Calculate the unit pulse response of the system



$$y[n] = \frac{1}{2}x[n] + \frac{1}{2}x[n-1]$$

$$h[n] = \frac{1}{2}\delta[n] + \frac{1}{2}\delta[n-1]$$



6. Unit Impulse Response

Example 2

Find the unit pulse response of the **causal** system described by the following difference equation:

$$y[n] - 0.8y[n-1] = x[n]$$

$$h[n] - 0.8h[n-1] = \delta[n]$$

When $n > 0$

$$h[n] - 0.8h[n-1] = 0$$

Equation
becomes
homogeneous

Using the equivalent Initial condition method, the difference equation is regarded as a homogeneous equation $\delta(n)$ convert to the initial condition:

Zero-State condition

$$h[-1] = 0$$

Can iterate out $h[0]$

$$h[0] = 0.8h[-1] + \delta[0] = 1$$

6. Unit Impulse Response

Example 2

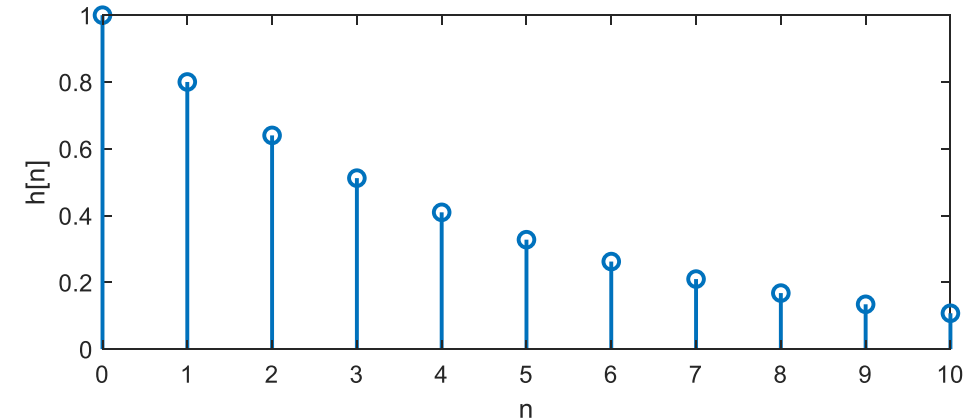
$$h[n] - 0.8h[n-1] = 0$$

Characteristic equation $r - 0.8 = 0$

Characteristic root $r = 0.8$

Unit pulse response

$$h[n] = C0.8^n$$



Substitute $h[0]=1$ into $h[n] = C0.8^n$, Get $C = 1$

$$\text{Thus } h[n] = 0.8^n u[n]$$

2. Judging Causality and Stability from $h[n]$

Causal system : output at any time n (or t) depends only on the current and past inputs, not future inputs

Necessary and sufficient conditions for a linear Time-invariant system to be a **causal system**:

$$n < 0 \quad h[n] = 0$$

The system meets the necessary and sufficient conditions for **stability**:

$$\sum_{n=-\infty}^{\infty} |h[n]| = P < \infty$$

The absolute sum of the unit impulse response is finite (**absolutely summable**)

Example 4

$$h[n] = a^n u[n]$$

(1) Causality:

One-sided, When $n < 0$, $h[n] = 0$

So, causal

(2) Stability:

$$\sum_{n=-\infty}^{\infty} |h[n]| = \sum_{n=0}^{\infty} |a^n| = \begin{cases} \frac{1}{1-|a|} & |a| < 1 \\ \infty & |a| \geq 1 \end{cases}$$
$$|a| < 1, \quad h[n]$$

When $|a| < 1$, the system is stable

6. Unit Impulse Response

Summary



The **zero-state response** of the system under the excitation of $\delta[n]$ is expressed as $h[n]$

$$h[-n] = 0 \quad n = 1, 2, 3, \dots, N$$

Necessary and sufficient condition for a linear Time-invariant system to be **a causal system**:

$$n < 0 \quad h[n] = 0$$

Necessary and sufficient conditions for **stability**:

$$\sum_{n=-\infty}^{\infty} |h[n]| = P < \infty$$

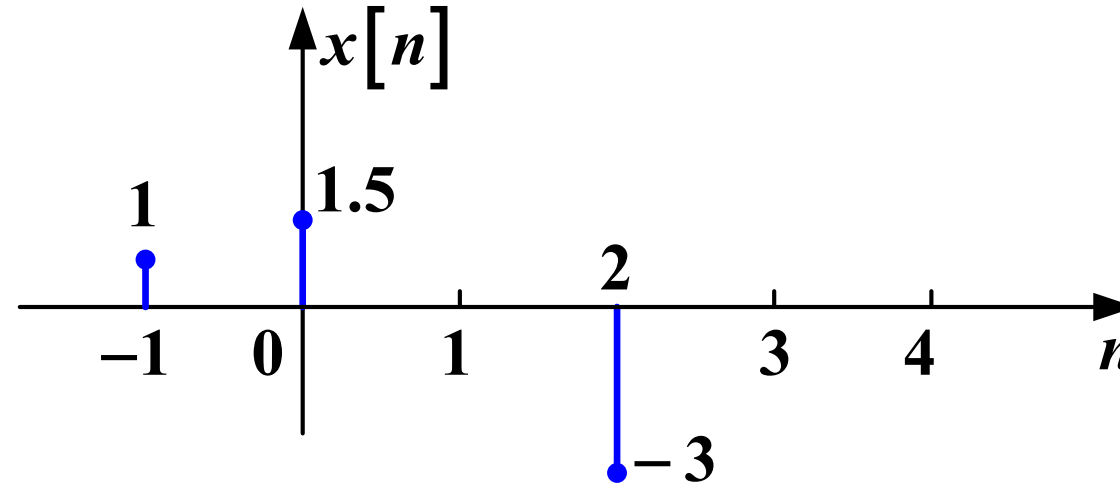
Main Content

- Definition of Convolution Sum
- Properties of Discrete Convolution
- Computation of Convolution Sum

1. Definition of Convolution Sum

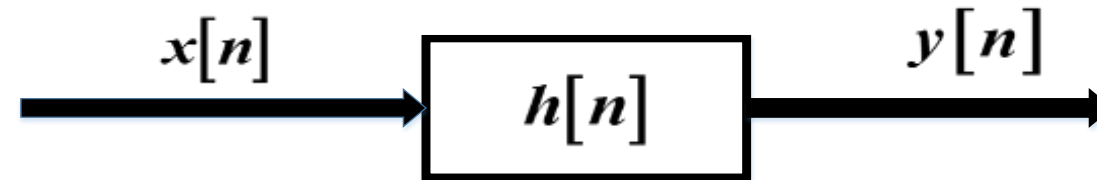
Any sequence $x[n]$ can be represented as a linear combination of weighted, shifted $\delta[n]$

$$x[n] = \sum_{m=-\infty}^{\infty} x[m] \delta[n-m]$$



$$x[n] = \delta[n+1] + 1.5\delta[n] - 3\delta[n-2]$$

1. Definition of Convolution Sum



Known $\delta[n] \rightarrow h[n]$

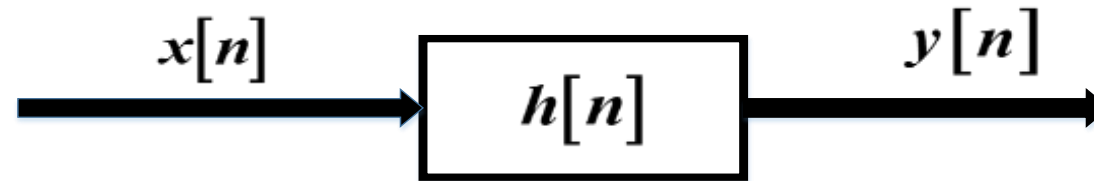
Time Invariant $\delta[n-m] \rightarrow h[n-m]$

Homogeneity $x[m]\delta[n-m] \rightarrow x[m]h[n-m]$

Additivity $x[n] = \sum_{m=-\infty}^{\infty} x[m]\delta[n-m]$

Output $y[n] = \sum_{m=-\infty}^{\infty} x[m]h[n-m] = x[n] * h[n]$

1. Definition of Convolution Sum



$$y[n] = \sum_{m=-\infty}^{\infty} \textcolor{red}{x}[m] \textcolor{blue}{h}[n-m]$$
$$= \textcolor{red}{x}[n] * \textcolor{red}{h}[n]$$

$\textcolor{blue}{h}[n]$, linking input and output, i.e. **zero state response**

$$y[n] = x[n] * h[n]$$

2. Properties of Discrete Convolution

1. Commutative property

$$x[n] * h[n] = h[n] * x[n]$$

2. Associative property

$$x[n] * h_1[n] * h_2[n] = x[n] * [h_1[n] * h_2[n]]$$

3. Distributive property

$$x[n] * [h_1[n] + h_2[n]] = x[n] * h_1[n] + x[n] * h_2[n]$$

4. Convolution with impulse sequences

$$x[n] * \delta[n - m] = x[n - m]$$

3. Computation of Convolution Sum

$$x[n] * h[n] = \sum_{m=-\infty}^{\infty} x[m] h[n-m]$$

The output range m is co-determined by $x[n]$ and $h[n]$

Discrete convolution process: sequence inversion → shift

→ multiplication → sum

(1) Analytical Method

(2) Graphical Method

(3) Use Properties

(4) Sum of Pairwise Multiplication

Example 1: Analytical Method

Given $h[n] = \alpha^n u[n]$ ($0 < \alpha < 1$), find the unit step response of the system

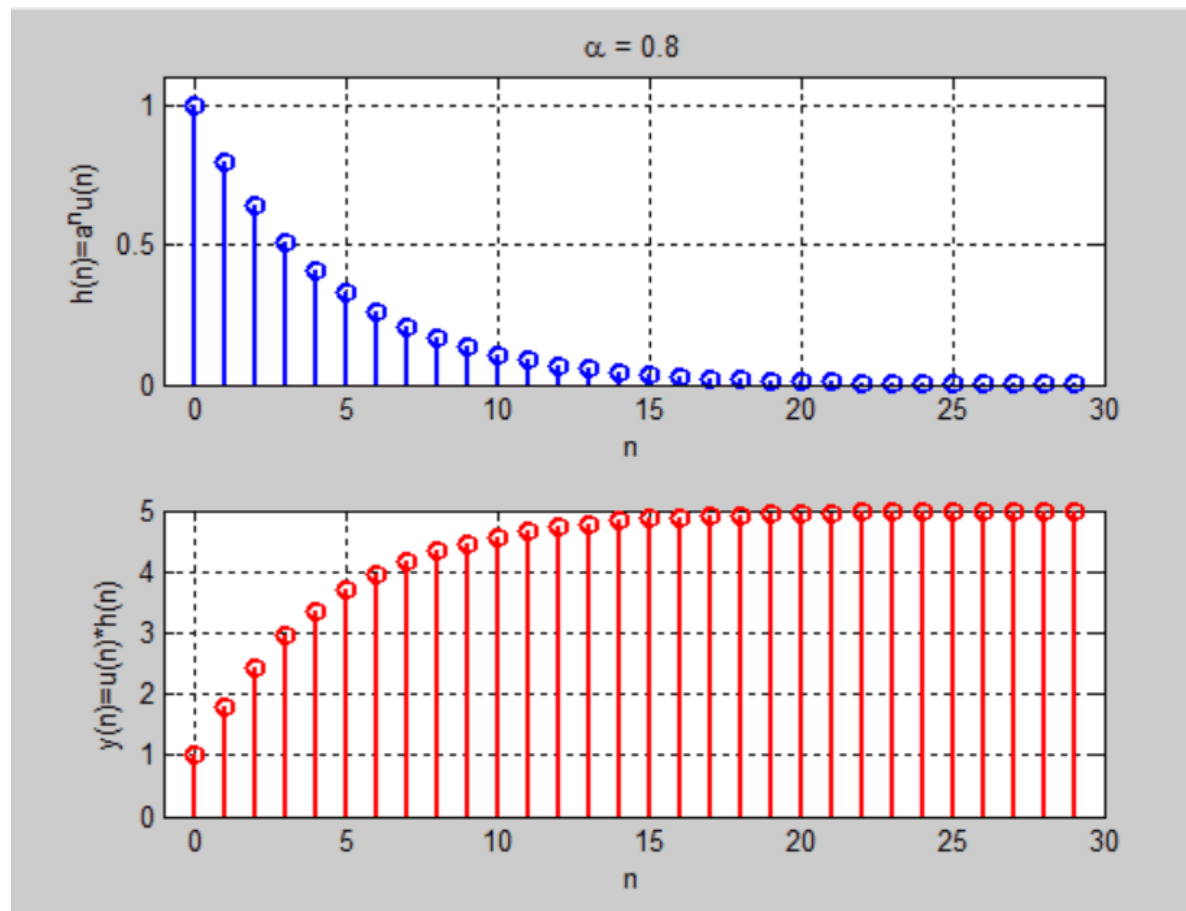
$$\begin{aligned} y[n] &= h[n] * u[n] \\ &= \sum_{m=-\infty}^{\infty} \alpha^m u[m] u[n-m] \quad \text{Argument : } \begin{cases} m \geq 0 \\ n-m \geq 0 \end{cases} \\ &= \left(\sum_{m=0}^n \alpha^m \right) \cdot u[n] \quad \Rightarrow \begin{cases} 0 \leq m \leq n \\ n \geq 0 \end{cases} \\ &= \frac{1 - \alpha^{n+1}}{1 - \alpha} u[n] \end{aligned}$$

7. Discrete Convolution

Example 1: Analytical Method

$$h[n] = \alpha^n u[n] \quad (0 < \alpha < 1) \text{ Input } u[n]$$

$$y[n] = \frac{1 - \alpha^{n+1}}{1 - \alpha} u[n]$$



Example 2: Graphical Method, Multiply and Sum

Consider the following finite length signal

$$x[n] = \begin{cases} 1, & 0 \leq n \leq 4 \\ 0, & \text{other } n \end{cases}$$

Calculate $y[n] = x[n] * x[n]$ using analytical methods

Analysis:

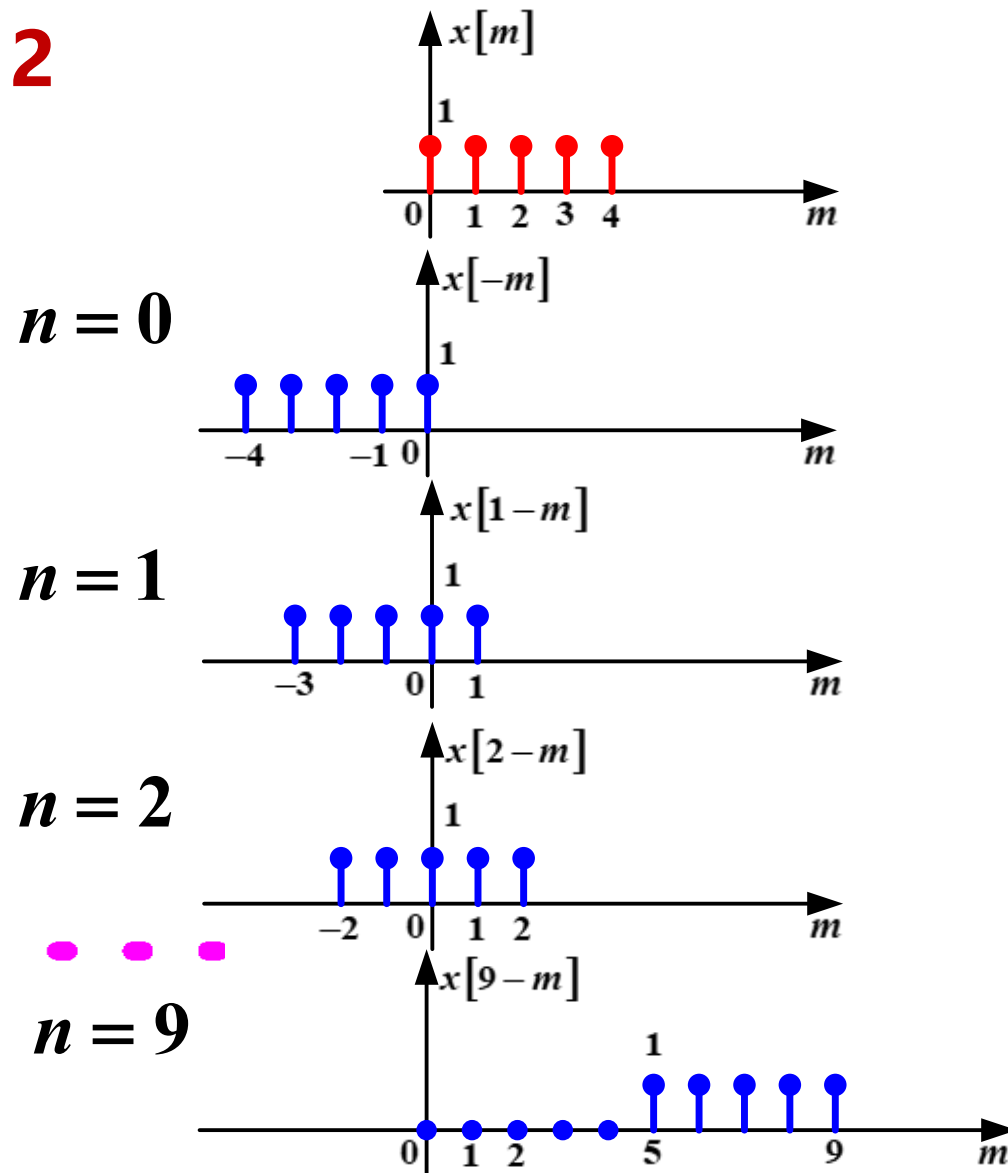
$$x[n] * h[n] = \sum_{m=-\infty}^{\infty} x[m] h[n-m]$$

Discrete convolution process: sequence inversion → shift → multiplication → sum

The m range is co-determined by $x[n]$ and $h[n]$

7. Discrete Convolution

Example 2



$$y[0] = 1 \times 1 = 1$$

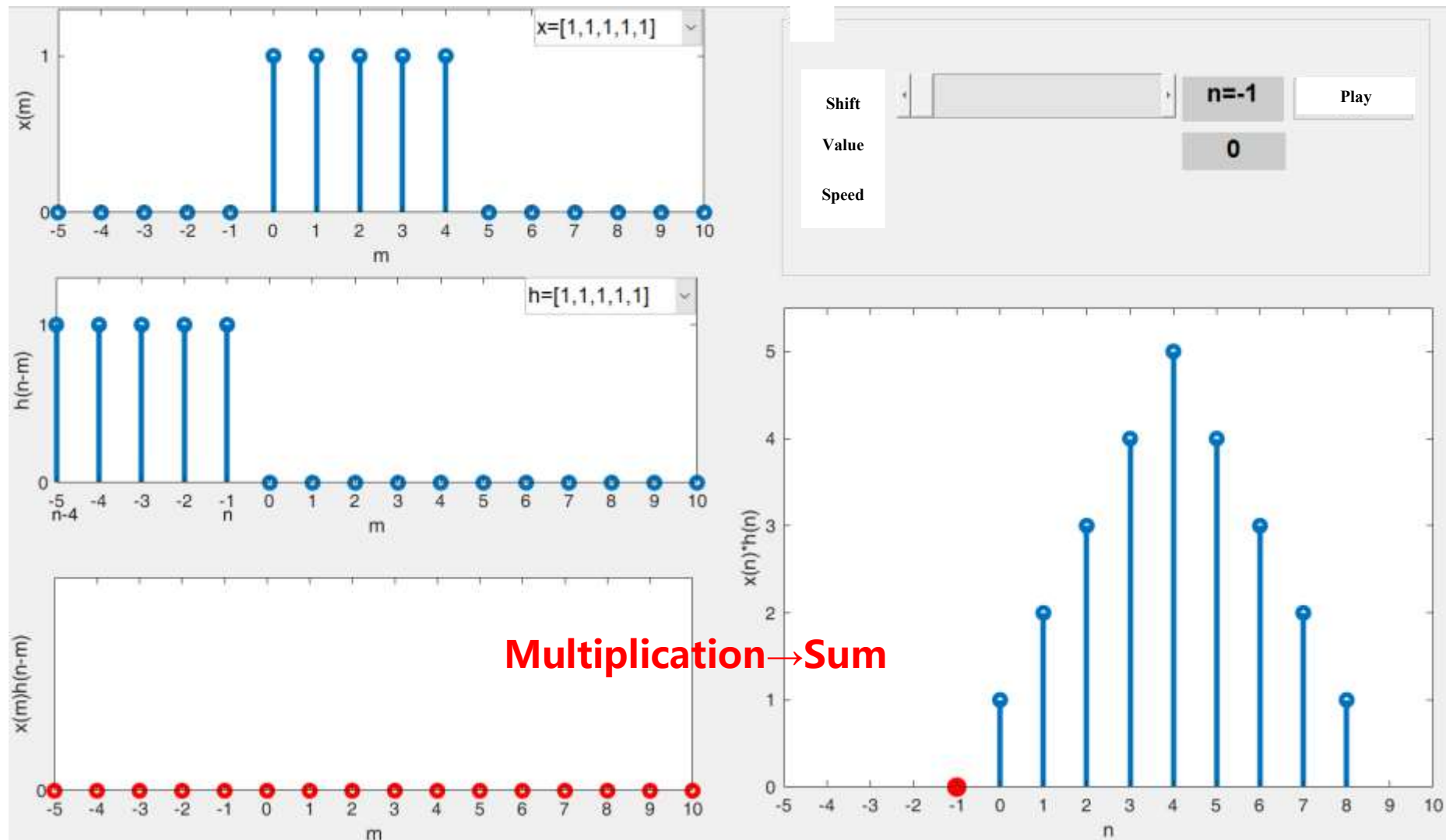
$$y[1] = 1 \times 1 + 1 \times 1 = 2$$

$$y[2] = 3$$

$$y[9] = 0$$

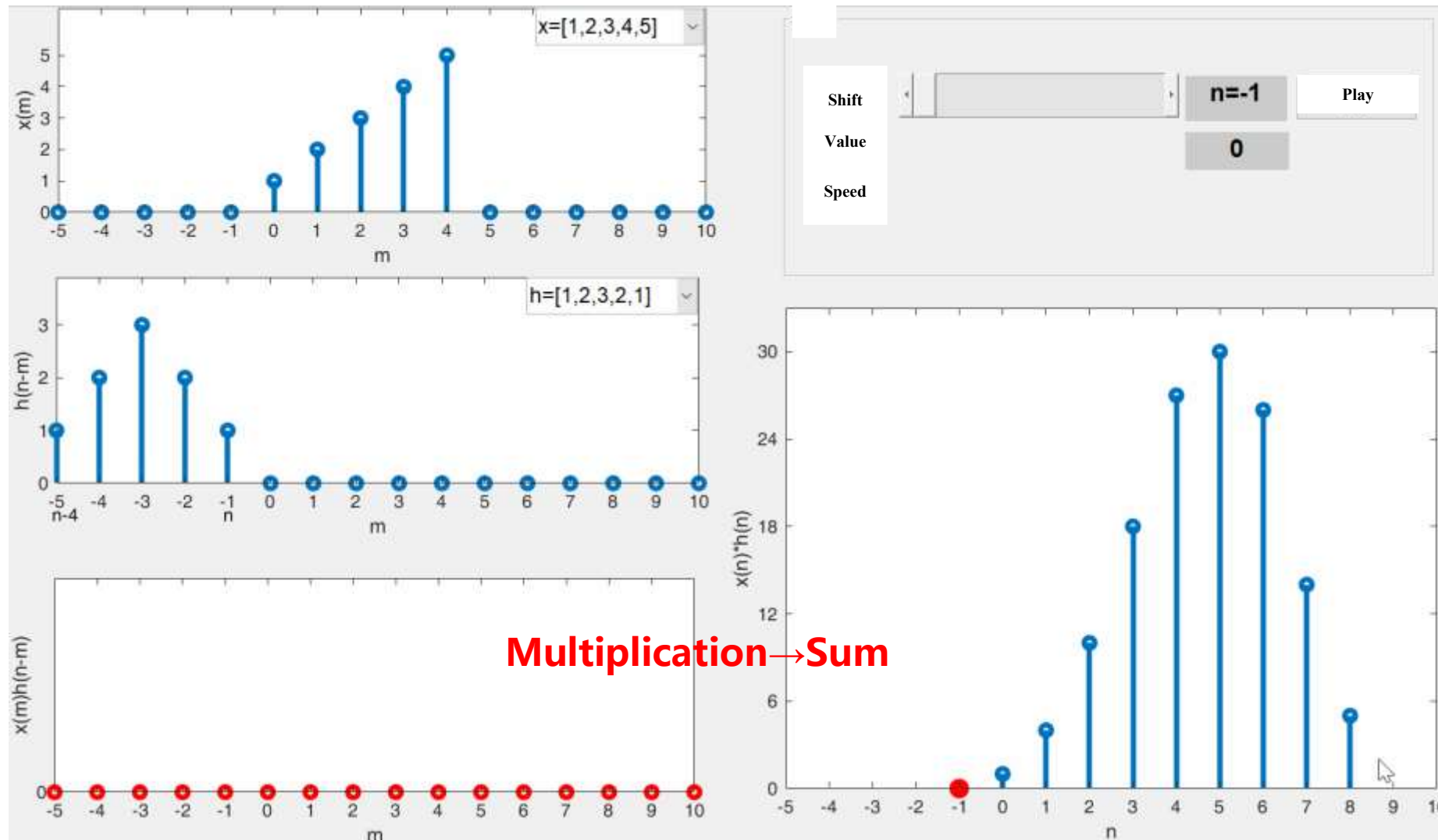
7. Discrete Convolution

Example 2: Graphical Method, Multiply and Sum



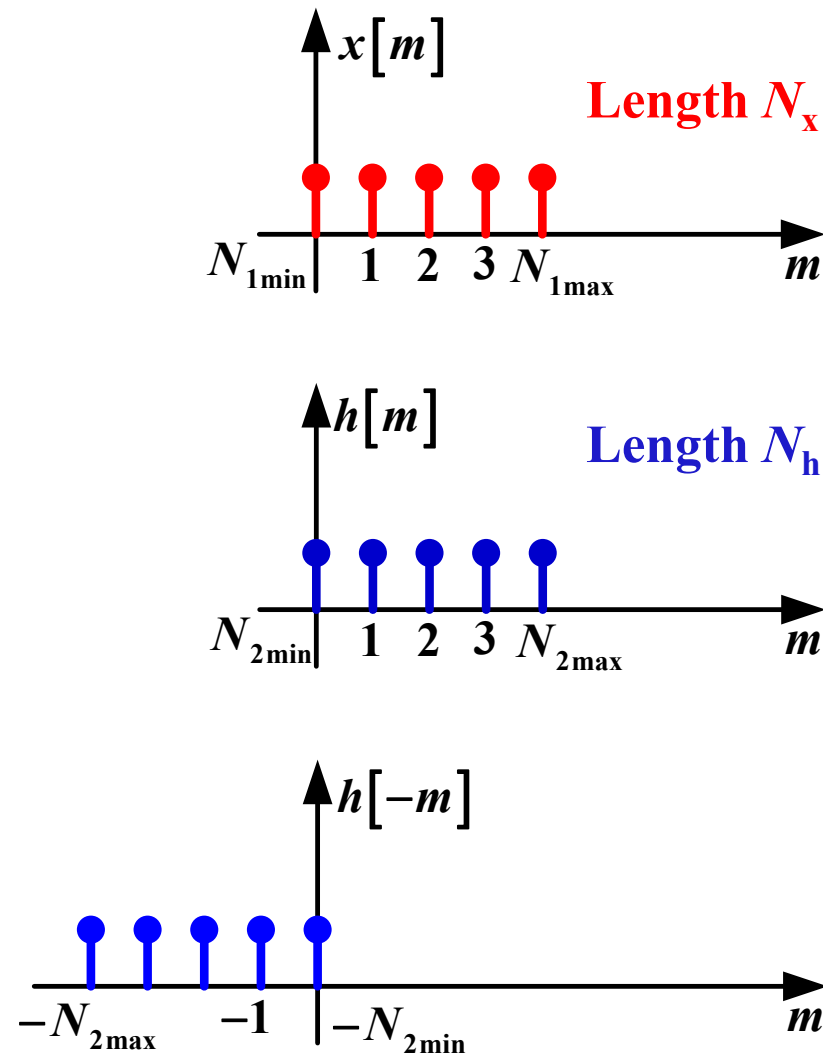
7. Discrete Convolution

Example 2: Graphical Method, Multiply and Sum



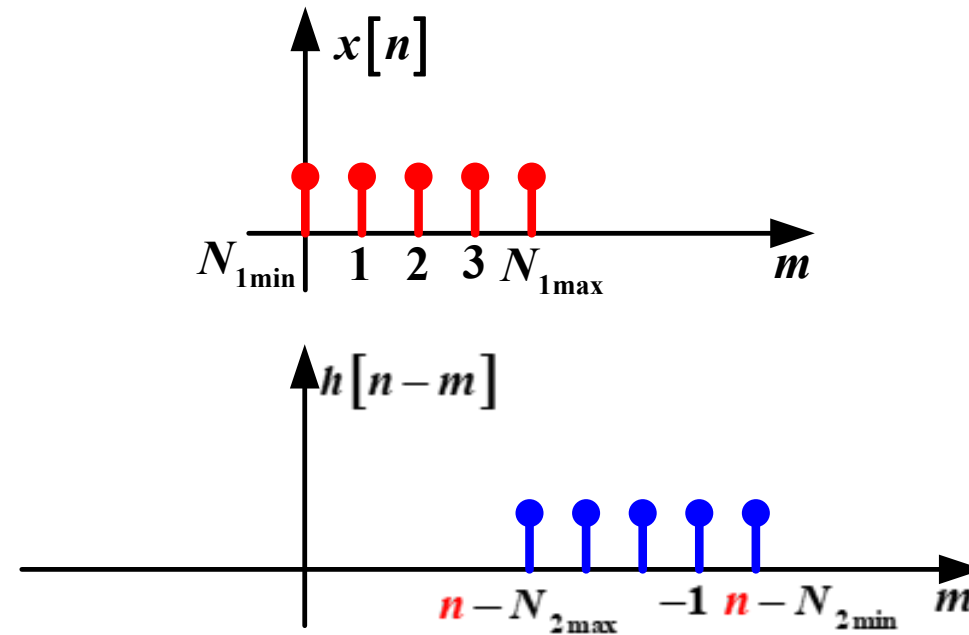
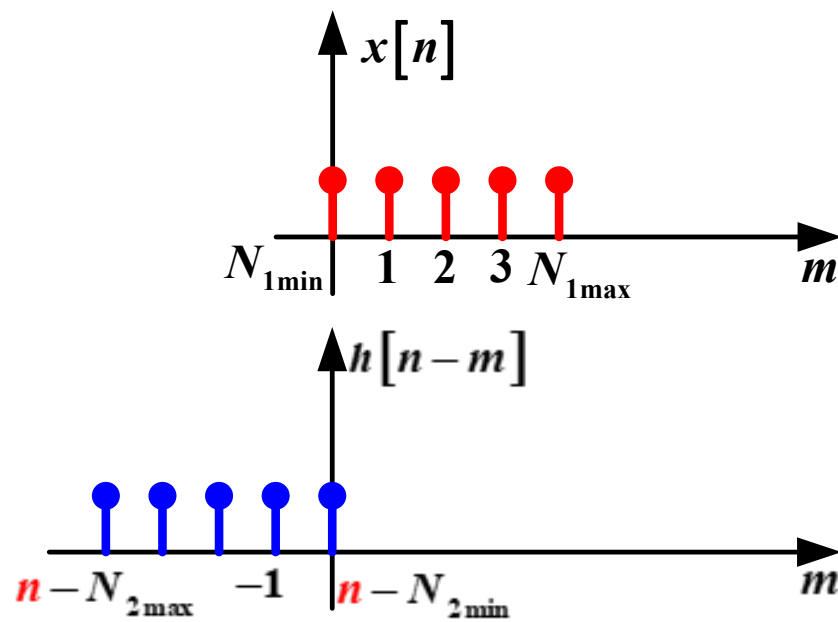
7. Discrete Convolution

Determining the Range of Convolution Results



7. Discrete Convolution

Determining the Range of Convolution Results



Lower limit 1 + Lower limit 2 $\leq n \leq$ Upper limit 1 + Upper limit 2

The length of convolutional signals $N_x + N_h - 1$

Example 3: Use Properties

$$\text{Given } x[n] = R_3[n], h[n] = \left\{ \underset{\substack{\uparrow \\ n=0}}{1}, 2, 3 \right\}, \text{ Get } x[n] * h[n].$$

Analysis:

$$x[n] * \delta[n - m] = x[n - m]$$

$$x[n] = \delta[n] + \delta[n - 1] + \delta[n - 2]$$

$$h[n] = \delta[n] + 2\delta[n - 1] + 3\delta[n - 2]$$

Example 3: Use Properties

$$x[n] = \delta[n] + \delta[n-1] + \delta[n-2]$$

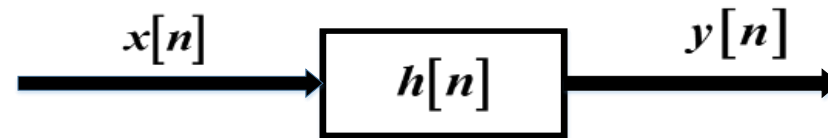
$$h[n] = \delta[n] + 2\delta[n-1] + 3\delta[n-2]$$

Distributive property

$$\begin{aligned} x[n] * h[n] &= \delta[n] + 2\delta[n-1] + 3\delta[n-2] \\ &\quad + \delta[n-1] + 2\delta[n-2] + 3\delta[n-3] \\ &\quad + \delta[n-2] + 2\delta[n-3] + 3\delta[n-4] \\ &= \delta[n] + 3\delta[n-1] + 6\delta[n-2] + 5\delta[n-3] + 3\delta[n-4] \end{aligned}$$

Summary

- Definition of Convolution Sum
- Properties of Discrete Convolution
- Computation of Convolution Sum



$$y[n] = \sum_{m=-\infty}^{\infty} \textcolor{red}{x}[m] \textcolor{blue}{h}[n-m] = x[n] * \textcolor{red}{h}[n]$$

- (1) Analytical Method
- (2) Graphical Method
- (3) Use Properties
- (4) Sum of Pairwise Multiplication



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Thank you for listening !

